Corporate Debt Maturity and Business Cycle Fluctuations

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Long-term debt is the main source of firm-financing in the U.S. We show that accounting for debt maturity is crucial for understanding business cycle dynamics. We develop a macroeconomic model with defaultable long-term debt and costly equity issuance. With long-term debt, firms have an incentive to increase leverage in order to dilute the value of outstanding debt. When equity issuance is costly, this incentive helps firms raise more debt through a debt dilution channel and mitigates the decline in net worth through a balance sheet channel, dampening the fall in investment in response to a negative financial shock. Using firm-level data, we estimate equity issuance costs and incorporate these estimates into a medium-scale DSGE model. We find that, once debt maturity and costly equity adjustment are accounted for, credit supply shocks emerge as the primary driver of business cycle fluctuations. In an extension with financially constrained intermediaries, these credit supply shocks can be interpreted as shocks to bank net worth.
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Figure 1 shows that the vast majority of U.S. non-financial corporate debt is long-term. The strong reliance on long-term financing raises important questions about the role of firms' debt maturity in determining the sensitivity of the economy to shocks.

Traditional macroeconomic models of firm financing, such as Bernanke, Gertler and Gilchrist (1999) and Kiyotaki and Moore (1997), focus on how financial frictions can amplify the effects of monetary policy and other shocks. As investment and the price of capital decline, the balance sheets of borrowers deteriorate, financial constraints tighten, and investment falls further through a *financial accelerator* channel. These models focus on short-term debt and do not allow firms to issue new equity.

On the other hand, the corporate finance literature has studied how long-term debt can generate under-investment through a *debt overhang* channel (Myers, 1977; Hennessy, 2004): Equity investors inject less capital when the returns on

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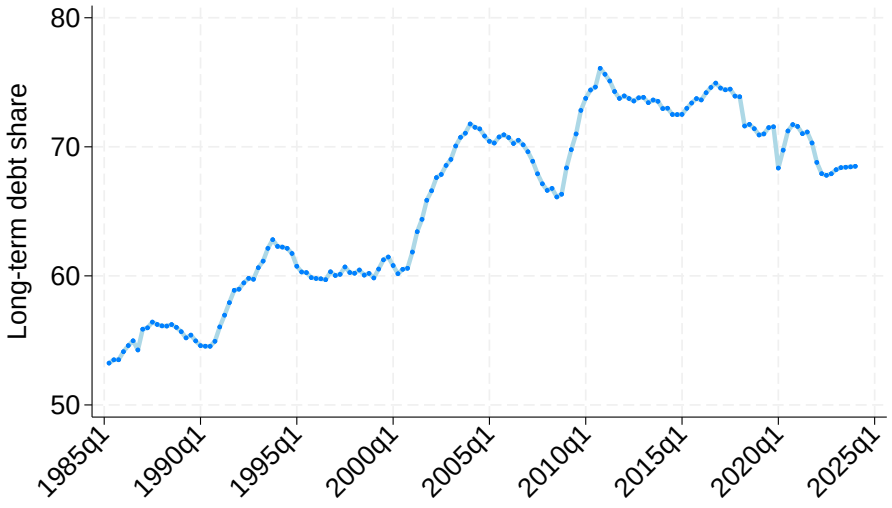


FIGURE 1. LONG-TERM DEBT SHARE

Note: Figure shows the share of debt maturing in more than one year as a percentage of total debt for non-financial corporate businesses. Source: U.S. Financial Accounts (Board of Governors of the Federal Reserve System, 2026*a*).

investment primarily benefit existing creditors. Gomes, Jermann and Schmid (2016) incorporate nominal, defaultable long-term debt into a macroeconomic model and show how this channel can amplify aggregate fluctuations. The presence of outstanding long-term debt creates a commitment problem for the firm which generates an incentive to increase leverage in order to dilute the value of outstanding debt. This mechanism leads to persistently elevated default rates, lower equity financing, and a stronger contraction in investment in response to standard macroeconomic shocks. However, because Gomes, Jermann and Schmid (2016) assume that firms can issue equity without costs, they do not allow for a financial accelerator channel.

Our paper bridges the gap between these two strands of literature by developing a model in which firms finance investment with defaultable long-term debt *and* costly equity issuance. We use this model to make three contributions. First, we show that equity adjustment costs are key in determining the role of long-term debt in transmitting shocks: When equity adjustments are costless, long-term debt amplifies the response of the economy to shocks, whereas the effect of shocks is dampened when it is costly to adjust equity. Second, using balance sheet data for U.S. firms, we estimate equity adjustment costs to be substantial. Third, we embed our mechanism into an estimated medium-scale DSGE model and show that including the observed maturity structure of corporate debt together with realistic equity adjustment costs leads us to reassess the drivers of business cycle

fluctuations. In particular, we find that credit supply shocks are the main driver of business cycle fluctuations.

We now turn to a more detailed discussion of these contributions. In our framework, long-term debt and equity issuance costs interact to shape the response of the economy to shocks. We illustrate this point by considering two benchmarks that have been studied in the literature: one in which equity issuance is costless and one in which firms cannot issue equity at all. In line with Gomes, Jermann and Schmid (2016), we find that when equity issuance is costless, a shock that increases the real value of firms' debt leads to a decline in investment and output only when debt is long-term. The higher real burden of debt increases firms' incentive to dilute preexisting creditors by increasing leverage, resulting in higher default rates and lower investment through a *debt overhang* channel. When considering the same shock in an economy where firms cannot issue equity, we find that an increase in the real value of debt leads to a drop in net worth and in investment through a financial accelerator channel, both with short and long-term debt. However, long-term debt now plays a dampening role: the increase in leverage and the lower price of outstanding debt obligations, caused by the debt dilution motive, increase net worth and result in higher investment.

A novel contribution of our paper is to show that the same debt dilution incentive that is responsible for the drop in investment without equity adjustment costs, dampens the decline in investment when equity issuance is costly.

We use balance sheet data from Compustat (S&P Global Market Intelligence, 2025) together with the model's optimality conditions to estimate the costs of adjusting equity. In line with the evidence from the corporate finance literature (e.g., Hennessy and Whited (2007)) we find that these costs are large, suggesting that costly equity injections are important for shaping firms' investment decisions.

In a quantitative exercise, we embed our estimates into an estimated medium scale DSGE model with financial frictions similar to Christiano, Motto and Rostagno (2014) and find that shocks to the supply of credit emerge as the main driver of business cycle fluctuations. The credit supply shock is the only shock that induces data-consistent co-movements between macroeconomic and financial variables. In contrast, other prominent shocks from the literature, i.e., both shocks to the marginal efficiency of investment (Justiniano, Primiceri and Tambalotti, 2010) and risk shocks (Christiano, Motto and Rostagno, 2014) cause equity to increase in response to a recessionary shock. We show that the presence of long-term debt and costly equity issuance are two key ingredients for the inference regarding which shocks drive business cycle fluctuations. First, accounting for debt maturity is crucial for the risk shock to be driven out by the credit supply shock. With long-term debt, the risk shock increases firms' debt dilution incentives and causes the value of outstanding debt to decline. As a result, the price of capital declines by less than the price of debt, leading to a counterfactual increase in equity. On the other hand, with short-term debt an increase in idiosyncratic investment risk causes equity to decline because there is no valuation effect on

previously issued debt.[1] Second, accounting for the substantial costs of adjusting equity is necessary for the credit supply shock to generate a recession. Without equity adjustment costs, firms respond to a credit supply shock by raising additional capital while reducing debt. The decline in leverage reduces debt overhang and stimulates investment.

We extend our model to include balance sheet constraints on financial intermediaries and find that the dampening effect of long term debt is still present. We use this model to provide an alternative interpretation of exogenous credit supply shocks as shocks to intermediaries' net worth. We show that the macroeconomic effects of shocks to credit supply in our baseline model are nearly identical to the effects of shocks to financial intermediaries' net worth in our extended model.

An additional contribution of our paper is methodological. The solution of models with long-term defaultable debt is complicated by the need to compute the derivative of the leverage policy function, which cannot be obtained with standard perturbation methods. We overcome this challenge by developing a novel algorithm which is based on a global solution of the problem at the steady state and a linear approximation of the derivative of the policy function with respect to the aggregate macro variables. This strategy makes Bayesian estimation of our DSGE model feasible.

Related literature. — By studying how the interaction of long-term corporate debt and financial frictions shapes the response to macroeconomic shocks in a medium-scale DSGE model, our work contributes to three strands of literature.

First, a large literature has studied the implications of corporate debt for macroeconomic fluctuations. These models typically focus on one-period debt and thus abstract from any frictions related to debt with long maturity (e.g. Bernanke, Gertler and Gilchrist (1999), Kiyotaki and Moore (1997), Christiano, Motto and Rostagno (2014), Arellano, Bai and Kehoe (2019)). Furthermore, firms typically cannot raise equity in these models.[2] Papers in this literature include a financial accelerator mechanism which links firm investment to changes in net worth, but abstract from the interaction between the value of long-term debt and firms' balance sheets.[3]

Second, we contribute to the literature on the effects of defaultable long-term corporate debt. Miao and Wang (2010) incorporate long-term defaultable corporate bonds and credit risk in a dynamic stochastic general equilibrium business

[1] If we assume one-period debt - ignoring the actual maturity structure of corporate debt in the data - we find that risk shocks are the main driver of fluctuations in investment, in line with Christiano, Motto and Rostagno (2014).

[2] Notable exceptions include Jermann and Quadrini (2012), Gertler, Kiyotaki and Queralto (2012), and Gertler, Kiyotaki and Prestipino (2020*a*), which feature short-term debt and (costly) equity adjustments. Ferrante (2019) features long-term debt but assumes infinitely costly equity issuance.

[3] A related literature focuses on firm heterogeneity. Jungherr et al. (2024) show that investment of firms with a larger amount of maturing debt is more responsive to monetary policy and rationalize this finding with a heterogeneous firm model with defaultable long-term debt and costly equity adjustments. Brunnermeier et al. (2023) show that firms with a larger share of long-term debt were more resilient during the German hyperinflation period of 1919-1923.

cycle model and show that credit risk amplifies aggregate fluctuations. Gomes, Jermann and Schmid (2016) show that unanticipated changes in inflation can have persistent effects on aggregate investment by affecting the real burden of long-term nominal debt. Relatedly, Jungherr and Schott (2022) show that risky long-term debt at the firm-level can lead to an amplified investment response to technology shocks and thereby rationalize the slow adjustment of aggregate leverage during recessions. These papers highlight the distortionary effect of the debt overhang channel on firms' investment decisions but abstract from equity issuance costs. Our paper instead focuses on the interplay between long-term debt and firms' net worth when equity is costly to adjust.[4,5]

A third strand of literature investigates the drivers of U.S. business cycle fluctuations in macroeconomic models. In their seminal contribution Smets and Wouters (2007) found that demand shocks accounted for the bulk of short-run output fluctuations, whereas supply shocks played an important role for output variations in the medium- to long-run. Justiniano, Primiceri and Tambalotti (2010) found shocks to the marginal efficiency of investment to be the driving force of the business cycle. Christiano, Motto and Rostagno (2014) revisit this result in a model with financial frictions. They show that once financial variables are included in the estimation, a risk shock that affects the volatility of idiosyncratic uncertainty explains most of the variation in output. We contribute to this literature by studying the important role played by a credit supply shock in accounting for business cycle fluctuations. Our results are consistent with work by Justiniano, Primiceri and Tambalotti (2019) and Gertler, Kiyotaki and Prestipino (2020*b*) who emphasize the role of credit supply shocks in explaining the Great Recession.

The structure of the paper is as follows. Section 2 describes the model. Section 3 explains the interaction between equity adjustment costs and debt maturity in determining the response of the economy to financial shocks. Section 4 contains our quantitative exercises. Section V extends the model to include financially constrained intermediaries. Section 6 concludes.

II. Model

There is a continuum of households, each consisting of a continuum of members. At each point in time a proportion e of household members are entrepreneurs, a proportion b are bankers, and the remaining proportion $1-e-b$ are workers. We

[4]The interaction between debt maturity and default has also been studied in the sovereign debt literature. See, for example, Hatchondo, Martinez and Sosa-Padilla (2016), Aguiar et al. (2019) and Bocola and Dovis (2019).

[5]Market participants may try to mitigate the commitment problem generated by risky long-term debt through contracting features such as debt covenants (e.g. Xiang, 2024). While covenants are pervasive in bank loans, they are relatively infrequent for investment grade corporate bonds, a major source of U.S. corporate debt. Further, covenants are frequently weakened by "fine print" clauses (Ivashina and Vallee, 2025), and in a majority of covenant violations, creditors take no action and there are no consequences for the borrowing firm (Roberts and Sufi, 2009). We therefore abstract from debt covenants in this paper.

describe the optimization problems of these three types of agents in turn.

A. Workers

Workers choose household consumption, C_t, banks deposits, D_t, and government bonds, B_t, to maximize

$$\sum_{t=0}^{\infty} \beta^t \left[\xi_t^c \log\left(C_t - hC_{t-1}\right) - \frac{\bar{\varphi}}{1+\varphi} \int h_{it}^{1+\varphi} di \right],$$

where ξ_t^c represents a preference shock. Workers' budget constraint is given by

$$C_t + D_t + B_t = \int w_{it} h_{it} di + D_{t-1} \frac{R_{t-1}^d}{\pi_t} + B_{t-1} \frac{R_{t-1}}{\pi_t} + T_t,$$

where R_{t-1}^d and R_{t-1} are the nominal rates of return on deposits and government bonds between time $t-1$ and t, π_t is the rate of inflation, and T_t collects all transfers to the household from firms, entrepreneurs, bankers, and the government. Workers take as given individual labor supply h_t^i and wages w_t^i, which are determined by labor agencies as described in section II.F.

Households' optimal holdings of deposits and government bonds satisfy

$$1 = \mathbb{E}_t \Lambda_{t+1} \frac{R_t^d}{\pi_{t+1}} = \mathbb{E}_t \Lambda_{t+1} \frac{R_t}{\pi_{t+1}}, \tag{1}$$

where $\Lambda_{t+1} = \beta \frac{U_c(t+1)}{U_c(t)}$ is the households stochastic discount factor with

$$U_c\left(t\right) = \frac{\xi_t^c}{C_t - hC_{t-1}} - \beta h \mathbb{E}_t \frac{\xi_{t+1}^c}{C_{t+1} - hC_t}.$$

B. Entrepreneurs

Entrepreneurs hold the productive capital in the economy. We assume that entrepreneurs rent their capital to intermediate goods producers at a market rental rate r_t^k. An entrepreneur's return on capital investment is given by

$$\xi_t R_t^k = \xi_t \frac{r_t^k + (1-\delta) Q_t^k}{Q_{t-1}^k}, \tag{2}$$

where Q_t^k is the price of capital, δ is the depreciation rate, and ξ_t is an idiosyncratic shock. We assume that $\log(\xi_t)$ is distributed as a Normal, i.e. $\log \xi_t \sim$

$\mathcal{N}\left(\frac{-\sigma_{t-1}^2}{2}, \sigma_{t-1}^2\right)$, with fluctuations in σ_t capturing risk shocks.[6]

Entrepreneurs finance capital purchases with equity, x_t, and by issuing long-term nominal debt l_t. Their flow budget constraint at time t is

$$Q_t^k k_t = x_t + Q_t^l l_t,$$

where Q_t^l is the price of debt.[7] Each unit of debt issued at $t-1$ pays a coupon c_l at time t, while the remaining portion $1-\lambda_l$ remains outstanding. The parameter λ_l thus controls the maturity of corporate debt. When $\lambda_l = 1$, debt is short term whereas when $\lambda_l < 1$ debt maturity is greater than a quarter. The entrepreneur can default on its debt obligations. In that case, the entrepreneur exits the market and its creditors take over the productive capital, subject to default costs, as discussed in section II.D below.

The net worth of an entrepreneur who owns k_{t-1} units of capital and has outstanding debt of l_{t-1} at time t is given by

$$n_t = \xi_t R_t^k Q_{t-1}^k k_{t-1} - \frac{c_l + (1-\lambda_l) Q_t^l}{\pi_t} l_{t-1}. \tag{3}$$

Each entrepreneur chooses a fraction $1-\omega_t$ of net worth to distribute to households as dividends. Equity is given by $x_t = \Psi(\omega_t) n_t < \omega_t n_t$, where $\Psi(\omega_t)$ includes equity injection costs:

$$\Psi(\omega_t) = (1-\bar{\psi})\omega_t - \frac{\psi}{2}(\omega_t - \bar{\omega})^2.$$

Here, $\bar{\psi} \geq 0$, $\psi \geq 0$, and $\bar{\omega}$ is a target equity injection rate.[8] This formulation of equity injection costs is a simple way to capture agency problems that limit entrepreneurs' ability to substitute debt- with equity finance.[9] The parameter $\bar{\psi}$ represents a linear cost which guarantees that around the steady state of the model entrepreneurs have an incentive to use debt to finance capital purchases.[10] Without equity issuance costs, i.e., $\bar{\psi} = 0$ and $\psi = 0$, entrepreneurs are able to costlessly substitute debt- with equity finance in response to changing financial conditions, as is assumed, for instance, in Gomes, Jermann and Schmid (2016). At the other extreme, when no new equity can be issued, i.e., $\psi = \infty$, entrepreneurs always retain a fixed share of net worth $\bar{\omega}$ as equity, and their only active margin of financial adjustment is through debt. This is the assumption in most papers

[6] This shock was introduced by Williamson (1987) and its properties were studied more recently by Christiano, Motto and Rostagno (2014).

[7] For ease of notation, we are omitting the dependence of the price of debt on the entrepreneur's financial decisions. We will be explicit about it when describing the entrepreneur's optimality conditions.

[8] For simplicity, we assume that the equity injection costs are rebated lump sum to households.

[9] A similar approach to model costly dividend payouts is used in Jermann and Quadrini (2012). The costs are proportional to net worth in order to obtain aggregation of entrepreneurs' policy functions, as in Elenev, Landvoigt and Van Nieuwerburgh (2021).

[10] This cost plays a role similar to a tax advantage of debt, used, for example, in Gomes, Jermann and Schmid (2016).

with financial frictions following Bernanke, Gertler and Gilchrist (1999). We allow for ψ to take on any positive value and estimate its value in Section IV.B.

Let $V_t(k_{t-1}, l_{t-1}, \xi_t)$ be the value function of a non-defaulting entrepreneur that enters period t with capital k_{t-1}, debt l_{t-1}, and an idiosyncratic shock ξ_t. The value to the entrepreneur is given by the present discounted value of dividend payouts,

$$V_t\left(k_{t-1}, l_{t-1}, \xi_t\right) = \max_{k_t, l_t, \omega_t, n_t} \left(1-\omega_t\right) n_t + \mathbb{E}_t \Lambda_{t,t+1} \max\left\{0, V_{t+1}\left(k_t, l_t, \xi_{t+1}\right)\right\}, \tag{4}$$

subject to the definition of net worth (3) and the flow budget constraint

$$Q_t^k k_t = \Psi(\omega_t) n_t + Q_t^l l_t. \tag{5}$$

The value function is linear in k_{t-1}. To see this, let $\eta_t = \frac{l_t}{k_t}$ denote the entrepreneur's leverage ratio and define the per unit of capital value as $v_t(\eta_{t-1}, \xi_t) = \frac{V_t(k_{t-1}, l_{t-1}, \xi_t)}{k_{t-1}}$. Dividing equation (4) by k_{t-1} we get:

$$v_t\left(\eta_{t-1}, \xi_t\right) = \max_{\eta_t, \omega_t} \mu_t(\eta_t, \eta_{t-1}, \xi_t) \cdot \left[\left(1-\omega_t\right) + \frac{\Psi\left(\omega_t\right)}{Q_t^k - Q_t^l \eta_t} \mathbb{E}_t \Lambda_{t,t+1} \max\left\{0, v_{t+1}\left(\eta_t, \xi_{t+1}\right)\right\}\right], \tag{6}$$

where $\mu_t(\eta_t, \eta_{t-1}, \xi_t)$ is the entrepreneur's net worth per unit of capital, given by

$$\mu_t\left(\eta_t, \eta_{t-1}, \xi_t\right) \equiv \frac{n_t}{k_{t-1}} = \xi_t R_t^k Q_{t-1}^k - \frac{c_l + \left(1-\lambda_l\right) Q_t^l}{\pi_t} \eta_{t-1}, \tag{7}$$

and we are using equations (5) and (7) to get $\frac{k_t}{k_{t-1}} = \frac{\Psi(\omega_t)\mu_t}{Q_t^k - Q_t^l \eta_t}$. Note that with long term debt, i.e. $\lambda_l < 1$, an entrepreneur's net worth per unit of capital, μ_t, depends on their choice of leverage, η_t, through the effect of leverage on the price of debt, Q_t^l.[11]

From (6) and (7), the optimal value $v_t\left(\eta_{t-1}, \xi_t\right)$ is increasing in ξ_t. As a result, there exists a threshold value of the idiosyncratic shock to the return on capital, denoted as $\bar{\xi}_{t+1}$, such that an entrepreneur defaults when $\xi_{t+1} < \bar{\xi}_{t+1}$. The threshold value satisfies $v_{t+1}\left(\eta_t, \bar{\xi}_{t+1}\right) = 0$, which using (7), pins down $\bar{\xi}_{t+1}$ as the value of the idiosyncratic shock that makes net worth equal to zero:

$$\bar{\xi}_{t+1} = \frac{1}{R_{t+1}^k Q_t^k} \cdot \frac{c_l + \left(1-\lambda_l\right) Q_{t+1}^l}{\pi_{t+1}} \eta_t. \tag{8}$$

[11] By contrast, with short term debt, i.e. $\lambda_l = 1$, net worth per unit of capital does not depend on an entrepreneur's choices at time t. In this case, μ_t cancels out of the optimization problem.

EQUITY INJECTIONS. — The optimality condition for equity injections is given by the derivative of the value function in (6) with respect to ω_t:

$$\frac{1}{\Psi'(\omega_t)} = \frac{\mathbb{E}_t \Lambda_{t,t+1} \int_{\bar{\xi}_{t+1}}^{\infty} v_{t+1}(\eta_t, \xi_{t+1}) \, dF_t(\xi_{t+1})}{Q_t^k - Q_t^l \eta_t} = \gamma_t \tag{9}$$

It states that the marginal cost of equity, $\frac{1}{\Psi'(\omega_t)}$, is equal to the expected discounted return on equity injections, which we denote by γ_t. The term F_t is the distribution of idiosyncratic shocks at time $t+1$. To characterize the return γ_t further, it is useful to introduce the entrepreneur's Tobin's Q, i.e., the ratio between the value of a unit of net worth to the entrepreneur and its book value, $\varphi_t \equiv \frac{v_t}{\mu_t}$. Using (6) and (8) we can write φ_t as

$$\varphi_t = (1-\omega_t) + \Psi(\omega_t) \frac{Q_t^k}{Q_t^k - Q_t^l \eta_t^l} \mathbb{E}_t \Lambda_{t,t+1} \varphi_{t+1} \int_{\bar{\xi}_{t+1}}^{\infty} \left(\xi_{t+1} - \bar{\xi}_{t+1}\right) R_{t+1}^k dF_t(\xi_{t+1}). \tag{10}$$

Using $v_{t+1} = \varphi_{t+1}\mu_{t+1}$ together with the definition of μ_{t+1} from (7) to substitute for v_{t+1} in (9), we can express the expected discounted returns on equity investment as

$$\gamma_t = \frac{Q_t^k}{Q_t^k - Q_t^l \eta_t} \mathbb{E}_t \Lambda_{t,t+1} \varphi_{t+1} \int_{\bar{\xi}_{t+1}}^{\infty} \left(\xi_{t+1} - \bar{\xi}_{t+1}\right) R_{t+1}^k dF_t(\xi_{t+1}), \tag{11}$$

which says that one unit of equity gets leveraged into $\frac{Q_t^k}{Q_t^k - Q_t^l \eta_t}$ units of capital, providing a return $\xi_{t+1} R_{t+1}^k$ net of the debt repayment $\bar{\xi}_{t+1} R_{t+1}^k$, whenever the idiosyncratic shock is larger than the threshold. The optimality condition for equity injection in (9) then becomes

$$\frac{1}{\Psi'(\omega_t)} = \frac{Q_t^k}{Q_t^k - Q_t^l \eta_t} \mathbb{E}_t \Lambda_{t,t+1} \varphi_{t+1} \int_{\bar{\xi}_{t+1}}^{\infty} \left(\xi_{t+1} - \bar{\xi}_{t+1}\right) R_{t+1}^k dF_t(\xi_{t+1}). \tag{12}$$

LEVERAGE. — The optimality condition for leverage is given by

$$\frac{\partial \mu_t}{\partial \eta_t} \varphi_t + \frac{\Psi(\omega_t)\mu_t}{Q_t^k - Q_t^l \eta_t} \left[\left(Q_t^l + \frac{dQ_t^l}{d\eta_t} \eta_t \right) \gamma_t - \mathbb{E}_t \Lambda_{t,t+1} \int_{\bar{\xi}_{t+1}}^{\infty} \frac{\partial v_{t+1}}{\partial \eta_t} dF_t(\xi_{t+1}) \right] = 0, \tag{13}$$

The first term in (13) is given by

$$\frac{\partial \mu_t}{\partial \eta_t} = -\frac{(1-\lambda_l)\eta_{t-1}}{\pi_t} \frac{dQ_t^l}{d\eta_t}. \tag{14}$$

This term captures a debt dilution incentive that works through the negative effect of an increase in leverage on the price of pre-existing debt, $\frac{dQ_t^l}{d\eta_t} < 0$.[12] This incentive depends on the maturity of debt, λ_l, and disappears with short-term debt, i.e., $\lambda_l = 1$. In addition, the debt dilution incentive also increases with the marginal value of net worth, φ_t.

The second term, $\frac{\Psi(\omega_t)\mu_t}{Q_t^k - Q_t^l \eta_t^l}\left(Q_t^l + \frac{dQ_t^l}{d\eta_t}\eta_t^l\right)\gamma_t$, is the marginal benefit of increasing leverage associated with higher returns to capital, and the last term,

$$-\frac{\Psi\left(\omega_t\right)\mu_t}{Q_t^k - Q_t^l \eta_t^l}\mathbb{E}_t \Lambda_{t,t+1}\int_{\bar{\xi}_{t+1}}^{\infty}\frac{\partial v_{t+1}}{\partial \eta_t}dF_t\left(\xi_{t+1}\right),$$

captures the marginal cost of leverage. Using the envelope condition

$$\frac{dv_t}{d\eta_{t-1}} = -\frac{c_l + \left(1-\lambda_l\right)Q_t^l}{\pi_t}\varphi_t, \tag{15}$$

we can express the marginal cost of leverage as the cost of repaying the debt in case of non-default:

$$\mathbb{E}_t \Lambda_{t,t+1}\int_{\bar{\xi}_{t+1}}^{\infty}\frac{\partial v_{t+1}}{\partial \eta_t}dF_t\left(\xi_{t+1}\right) = \mathbb{E}_t \Lambda_{t,t+1} Q_t^l R_{t+1}^l \varphi_{t+1}\left(1 - F_t\left(\bar{\xi}_{t+1}\right)\right),$$

where R_{t+1}^l is the real return on debt of non-defaulting entrepreneurs:

$$R_{t+1}^l = \frac{1}{\pi_{t+1}}\cdot\frac{c_l + \left(1-\lambda_l\right)Q_{t+1}^l}{Q_t^l}. \tag{16}$$

Using (14) and (15) in (13) we can rewrite the optimality condition for entrepreneurial leverage as:

$$\mathbb{E}_t \Lambda_{t,t+1} R_{t+1}^l\left(1 - F_t\left(\bar{\xi}_{t+1}\right)\right)\varphi_{t+1} = \left(1+\epsilon_{\eta_t}\right)\gamma_t - \epsilon_{\eta_t}\frac{\left(1-\lambda_l\right)\eta_{t-1}}{\eta_t}\frac{\left(Q_t^k - Q_t^l\eta_t^l\right)}{\Psi\left(\omega_t\right)\mu_t}\frac{\varphi_t}{\pi_t}, \tag{17}$$

where $\epsilon_{\eta_t} = \frac{dQ_t^l}{d\eta_t}\frac{\eta_t}{Q_t^l} < 0$ is the elasticity of the price of debt with respect to leverage. Equations (8), (12) and (17), together with the definition of μ_t and φ_t in (7) and (10), are the optimality conditions for the entrepreneur's problem. We now turn to describing aggregation of the entrepreneurs' policy functions.

[12] Equations (24) and (26) report the pricing equation for entrepreneurial debt and its derivative. As we discuss below, $\frac{dQ_t^l}{d\eta_t} < 0$ in our model.

C. Aggregation of entrepreneurs' choices

Defaulting entrepreneurs exit and are replaced by an equal measure of new entrepreneurs. New entrants receive a transfer proportional to aggregate net worth, $T_t^e = \tau^e X_{t-1}$, and use it to purchase capital:

$$Q_t^k k_t^e = T_t^e + Q_t^l l_t^e \tag{18}$$

To preserve aggregation we make two assumptions. First we assume that new entrants' leverage is the same as the leverage of existing entrepreneurs, implying $l_t^e = \eta_t k_t^e$. Second, to ensure that net worth per unit of capital, μ_t in (17), is constant among surviving entrepreneurs, we assume that the ex-post return on capital is equal across non-defaulting entrepreneurs, implying constant leverage across all entrepreneurs active at time t.[13]

Aggregate equity of entrepreneurs at time t is then given by the sum of the equity of entrepreneurs that do not default at time t and the transfer to new entrepreneurs:

$$X_t = \Psi(\omega_t)\left(\tilde{\xi}_t - \bar{\xi}_t\right) R_t^k Q_{t-1}^k K_{t-1}\left(1 - F\left(\bar{\xi}_t\right)\right) + T_t^e + \xi_t^n, \tag{19}$$

with $\tilde{\xi}_t = \frac{1}{1-F_{t-1}\left(\bar{\xi}_t\right)} \int_{\bar{\xi}_t} \xi_t dF_{t-1}(\xi_t)$. The term ξ_t^n captures a net worth shock which exogenously affects aggregate entrepreneurial net worth.

Aggregate capital demand is given by

$$K_t = \frac{X_t}{Q_t^k - \eta_t Q_t^l}. \tag{20}$$

D. Financial intermediaries

Financial intermediaries raise deposits from households and invest these funds in corporate bonds according to

$$Q_t^l L_t = \kappa_t D_t, \tag{21}$$

where L_t and D_t respectively represent aggregate bonds and deposits, and κ_t is a credit supply shock that affects financial intermediaries' technology for transforming deposits into entrepreneurial debt.

[13] This is achieved with a state-contingent transfer to each non-defaulting entrepreneur, i.e. an entrepreneur with $\xi_{i,t} > \bar{\xi}_t$, given by $(\tilde{\xi}_t - \xi_{i,t}) R_t^k Q_t^k k_{t-1}$ where $\tilde{\xi}_t = \frac{1}{1-F_{t-1}\left(\bar{\xi}_t\right)} \int_{\bar{\xi}_t} \xi_t dF_{t-1}(\xi_t)$. This assumption allows us to abstract from idiosyncratic risk in the model solution, except for its effects on defaults. Technically, we are linearizing individual entrepreneurs' policy functions before aggregation and only keeping track of average values, thus avoiding to carry the entire distribution of leverage and idiosyncratic shocks across entrepreneurs.

The expected return on corporate bonds, R^b_{t+1}, is given by

$$R^b_{t+1} = \left(1 - F_t\left(\bar{\xi}_{t+1}\right)\right) R^l_{t+1} + \gamma_l \frac{Q^k_t R^k_{t+1}}{\eta_t Q^l_t} \int_0^{\bar{\xi}_{t+1}} \xi_{t+1} dF_t\left(\xi_{t+1}\right), \tag{22}$$

where the second term indicates that in case of default creditors only recover a fraction $\gamma_l < 1$ of the capital of defaulting entrepreneurs. Because each creditor lends to a continuum of entrepreneurs, R^b_{t+1} also represents the realized return on each creditors' debt portfolio.

Financial intermediaries are owned by households and their objective is to maximize profits

$$\max_{D_t} D_t \cdot \mathbb{E}_t \Lambda_{t,t+1} \left(\kappa_t R^b_{t+1} - \frac{R^d_t}{\pi_{t+1}} \right).$$

Optimality implies

$$\mathbb{E}_t \Lambda_{t,t+1} \kappa_t R^b_{t+1} = \mathbb{E}_t \Lambda_{t,t+1} \frac{R^d_t}{\pi_{t+1}} = 1, \tag{23}$$

where the second equality comes from households' first order conditions. Multiplying (23) by Q^l_t we obtain an expression for entrepreneurs' price of debt:

$$Q^l_t = \mathbb{E}_t \Lambda_{t+1} \kappa_t \left\{ \left(1 - F_t\left(\bar{\xi}_{t+1}\right)\right) \frac{c_l + (1-\lambda_l) Q^l_{t+1}}{\pi_{t+1}} + \gamma_l \frac{Q^k_t R^k_{t+1}}{\eta_t} \int_0^{\bar{\xi}_{t+1}} \xi_{t+1} dF_t\left(\xi_{t+1}\right) \right\} \tag{24}$$

Equations (23) and (24) show that the credit supply shock κ_t introduces an exogenous wedge between the expected return on bonds and the risk-free rate. A lower κ_t increases the lenders' required rate of return on corporate bonds, reducing the supply of credit and putting downward pressure on Q^l_t.

We can decompose the bond spread, i.e. the difference between the rate of return on bonds of non-defaulting entrepreneurs and the deposit rate, into a default premium and a liquidity premium:

$$\underbrace{\mathbb{E}_t(R^l_{t+1} - \frac{R^d_t}{\pi_{t+1}})}_{\text{Bond spread}} = \underbrace{\mathbb{E}_t(R^l_{t+1} - R^b_{t+1})}_{\text{Default premium}} + \underbrace{\mathbb{E}_t(R^b_{t+1} - \frac{R^d_t}{\pi_{t+1}})}_{\text{Liquidity premium}}. \tag{25}$$

The first component captures the expected default risk, which moves with aggregate shocks and is particularly sensitive to the risk shock σ_t, whereas the second component moves inversely with κ_t.[14]

Finally, using equation (24) we can compute the derivative of the debt price with respect to entrepreneurial leverage, which enters the first-order condition for

[14] We endogenize the liquidity premium by introducing an agency problem in the financial sector in Section V.

leverage in (17) as

$$\frac{\partial Q_t^l}{\partial \eta_t} = -\mathbb{E}_t \Lambda_{t+1} \kappa_t \left\{ \frac{\partial \bar{\xi}_{t+1}}{\partial \eta_t} f_t\left(\bar{\xi}_{t+1}\right) \left[Q_t^l R_{t+1}^l - \gamma_l \frac{Q_t^k R_{t+1}^k}{\eta_t} \bar{\xi}_{t+1} \right] + \gamma_l \frac{Q_t^k R_{t+1}^k}{\left(\eta_t\right)^2} \int_0^{\bar{\xi}_{t+1}} \xi_{t+1} dF_t\left(\xi_{t+1}\right) \right\}$$

$$+ \mathbb{E}_t \Lambda_{t+1} \kappa_t [1 - F_t\left(\bar{\xi}_{t+1}\right)] \left(1 - \lambda_l\right) \frac{\partial Q_{t+1}^l}{\partial \eta_{t+1}} \eta_{\eta,t+1}\left(\eta_t\right). \tag{26}$$

The term in the first row of (26) includes the impact of entrepreneurs' leverage on next period's default threshold and on the expected recovery rate in case of default. Because a higher η_t results in a higher expected probability of default and in a lower recovery rate, this term is negative. The term on the second row of (26) captures the effect of current leverage on the choice of future leverage, as measured by the derivative of the entrepreneurs' leverage policy function $\eta_{\eta,t+1}\left(\eta_t\right) = \frac{\partial \eta_{t+1}}{\partial \eta_t}$.[15] As shown in (17), and as noted by Gomes, Jermann and Schmid (2016), long-term debt introduces an incentive for entrepreneurs to dilute the value of preexisting debt by increasing leverage, and rational lenders take this effect into account. Intuitively, and as we show computationally below, the derivative $\frac{\partial \eta_{t+1}}{\partial \eta_t}$ is positive because higher initial leverage increases the benefit of diluting existing debt. Given $\frac{\partial \eta_{t+1}}{\partial \eta_t} > 0$, equation (26) implies that $\frac{\partial Q_t^l}{\partial \eta_t}$ is the present discounted sum of negative terms and is hence negative.

The computation of this derivative complicates the numerical solution of the model, because it cannot be computed with standard perturbation methods. As discussed below, to overcome this problem we develop an algorithm which uses global solution techniques to capture the local dynamics of this derivative around the steady state of the model.

E. Production

The final good Y_t is a CES composite of different intermediate varieties $Y_t\left(i\right)$ given by

$$Y_t = \left[\int Y_t\left(i\right)^{\frac{\varepsilon_t - 1}{\varepsilon_t}} \right]^{\frac{\varepsilon_t}{\varepsilon_t - 1}}, \tag{27}$$

where ε_t is time-varying due to a markup shock.

Intermediate varieties are sold by monopolistically competitive retailers who purchase a homogeneous good Y_t^m at price p_t^m, differentiate it into $Y_t(i)$, and sell it to the final good producers. The retailers face Rotemberg adjustment costs for deviating from a target inflation rate $\tilde{\pi}_t$, according to $\frac{1}{2} \cdot \kappa(\frac{\pi_t}{\tilde{\pi}_t} - 1)^2 \cdot$

[15] We are abusing notation here since the policy function is the *individual* entrepreneur's policy function for leverage as a function of individual entrepreneur's leverage. See Appendix for details.

Y_t. In particular, $\tilde{\pi}_t = \bar{\pi}_t^{\iota_p} \pi_{t-1}^{(1-\iota_p)}$ where $\bar{\pi}_t$ represents a time-varying inflation target set by the central bank, and ι_p captures the degree of inflation indexation. Aggregating their optimality conditions, we obtain a standard Phillips Curve

$$\left(\frac{\pi_t}{\tilde{\pi}_t} - 1\right)\frac{\pi_t}{\tilde{\pi}_t} = \left[P_t^m \frac{\varepsilon_t}{(\varepsilon_t - 1)} - 1\right]\gamma_p + \beta \mathbb{E}_t \Lambda_{t,t+1}\left(\frac{\pi_{t+1}}{\tilde{\pi}_{t+1}} - 1\right)\frac{\pi_{t+1}}{\tilde{\pi}_{t+1}}\frac{Y_{t+1}}{Y_t} \tag{28}$$

where $\gamma_p = (\varepsilon_t - 1)/\kappa$ represents the slope of the Phillips curve.

The homogeneous intermediate good is produced by a continuum of intermediate good producers. Intermediate good producer j hires labor at real wage w_t and rents capital at rate r_t^k to produce a homogeneous good Y_t^m using a Cobb-Douglas technology

$$Y_t^m(j) = A_t(u_t(j)K_{t-1}(j))^{\chi}(H_t(j))^{1-\chi}, \tag{29}$$

where A_t is aggregate TFP, and $H_t(j)$ and $K_{t-1}(j)$ denote labor and capital. Utilization u_t can be adjusted by paying a cost $a(u_t)K_{t-1} = \bar{\gamma}_u(e^{\gamma_u(u_t-1)}-1)K_{t-1}$, where, in the steady state, $a(1) = 0$ and $\bar{\gamma}_u$ is calibrated to obtain $a'(1) = r_{ss}^k$, the steady state rental rate on capital.

Finally, capital producers sell capital at price Q_t^k and face convex adjustment costs. They solve:

$$\max \mathbb{E}_t \sum_{i=0}^{\infty} \Lambda_{t,t+i}\left[Q_{t+i}I_{t+i} - \mu_t^k I_{t+i}\left(1 - \frac{\gamma_k}{2}\left(\frac{I_{t+i}}{I_{t+i-1}} - 1\right)^2\right)\right]$$

where μ_t^k captures a shock to the marginal efficiency of investment (MEI).

F. Labor agencies and labor unions

Labor agencies demand individual labor varieties from labor unions and aggregate them using a constant elasticity of substitution technology into a composite labor input that is sold to intermediate goods producers.

Optimal demand for labor h_{it} supplied by a labor union to the labor agency is:

$$h_{it} = \left(\frac{w_{it}}{w_t}\right)^{-\varepsilon_t^w} H_t, \tag{30}$$

where ε_t^w is a wage markup which is allowed to vary stochastically and w_t is the real wage paid by firms to purchase the labor composite H_t.

Labor unions choose wages and labor supply for individual labor varieties in order to maximize workers' utility. They face Rotemberg costs of adjusting wages $\frac{\kappa^w}{2}(\frac{\pi_t^w}{\tilde{\pi}_t^w} - 1)^2 H_t$, where $\pi_t^w = \frac{w_t}{w_{t-1}}\pi_t$ and $\tilde{\pi}_t^w = \bar{\pi}_t^{\iota_w}\pi_{t-1}^{(1-\iota_w)}$. The parameter ι_w captures the degree of wage indexation to inflation. Labor unions' optimization delivers the wage Phillips curve

(31)
$$\frac{\pi_t^w}{\tilde{\pi}_t^w}\left(\frac{\pi_t^w}{\tilde{\pi}_t^w}-1\right)=\left(\frac{\bar{\varphi}H_t^{\varphi}}{U_{ct}}\frac{\varepsilon_t^w}{\varepsilon_t^w-1}-W_t\right)\gamma_w+\beta\mathbb{E}_t\Lambda_{t,t+1}\frac{\pi_{t+1}^w}{\tilde{\pi}_{t+1}^w}\left(\frac{\pi_{t+1}^w}{\tilde{\pi}_{t+1}^w}-1\right)\frac{H_{t+1}}{H_t},$$

where $\gamma_w=\frac{\varepsilon_t^w-1}{\kappa^w}$ captures the slope of the wage Phillips curve.

G. *Monetary policy, market clearing, and shocks*

The central bank sets the interest rate on government bonds according to an inertial Taylor rule that responds to inflation (in deviation from a target) and output growth

(32)
$$\log(R_t)=(1-\rho_r)\,log(R_{SS})+\rho_r\log(R_{i,t-1})+(1-\rho_r)\left(\kappa_\pi\log\left(\frac{\pi_t}{\bar{\pi}_t}\right)+\kappa_y\log\left(\frac{Y_t}{Y_{t-1}}\right)\right)+\xi_t^m.$$

where ρ_r is a smoothing parameter and ξ_t^m is a monetary policy shock.

Market clearing in the goods market requires

$$Y_t-\nu\left[K_{t-1}(1-\gamma_l)\int_0^{\bar{\xi}_t}\xi_t dF_t(\xi_t)\right]=C_t+I_t+G_t+a(u_t)K_{t-1}, \tag{33}$$

where ν governs the share of default costs that translate into real resources losses, and G_t is an exogenous government spending shock.[16]

The model features 11 shocks: i) a monetary policy shock ξ_t^m; ii) a risk shock σ_t; iii) a net worth shock ξ_t^n; iv) a shock to the marginal efficiency of investment μ_t; v) a price markup shock ε_t; vi) a wage markup shock ε_t^w; vii) a TFP shock A_t; viii) a government spending shock G_t; ix) a shock to the inflation target $\bar{\pi}_t$; x) a credit supply shock κ_t; xi) a preference shock ξ_t^c. We assume that each shock s_t has an AR1 representation

$$s_t=\rho_s s_{t-1}+\zeta_t^s, \tag{34}$$

where $\zeta_t^s\sim N(0,\sigma^s)$ is an i.i.d. innovation and the parameter ρ_s captures the shock persistence. In addition, following Christiano, Motto and Rostagno (2014), we introduce a news component for the risk shock, by assuming that

$$\zeta_t^\sigma=\zeta_{0,t}^\sigma+\zeta_{1,t-1}^\sigma+\dots\zeta_{8,t-8}^\sigma, \tag{35}$$

where $\zeta_{0,t}^\sigma\sim N(0,\sigma^\sigma)$ represents the shock innovation realized at time t. The remaining terms are news shocks $\zeta_{1,t-1}^\sigma\dots\zeta_{8,t-8}^\sigma\sim N(0,\sigma_n^\sigma)$, known to the agents up to two years in advance. News shocks have a correlation structure described

[16] Appendix A collects agents' optimality conditions and all the equilibrium equations of the model.

by the parameter $\rho_{\sigma,n}$. Finally, we assume that the monetary policy shock and the net worth shock have an autocorrelation parameter equal to zero, that is $\rho_{\xi^n} = \rho_{\xi^m} = 0$.

III. Equity adjustment costs and long-term debt

One of our key findings is that equity adjustment costs play a crucial role for how the presence of long-term debt affects the transmission of shocks to the economy. To illustrate this result and to facilitate comparison with the existing literature, we study the effect of a negative inflation shock in a version of our model without nominal rigidities ($\kappa = \kappa_w = 0$) and without real costs of default, i.e., $\nu = 0$.[17] This shock is transmitted to the economy by increasing the real burden of corporate debt.[18] We present the response of the economy in *i)* a model without equity adjustment costs and *ii)* a model with infinite equity adjustment costs. In each of these models we compare the response of the economy with long-term debt to a model where debt is short-term.

A. *Model without equity adjustment costs*

We begin by discussing the effects of a shock to inflation in a model without equity adjustment costs, reminiscent of Proposition 1 in Gomes, Jermann and Schmid (2016).

When it is costless to adjust equity injections at the margin, i.e., $\psi = \bar{\psi} = 0$, the optimality conditions for the entrepreneur's problem deliver the three following aggregate equations:

$$1 = \frac{Q_t^k}{Q_t^k - Q_t^l \eta_t} \mathbb{E}_t \Lambda_{t,t+1} \int_{\bar{\xi}_{t+1}}^{\infty} \left(\xi_{t+1} - \bar{\xi}_{t+1}\right) R_{t+1}^k dF_t\left(\xi_{t+1}\right), \tag{36}$$

$$\mathbb{E}_t \Lambda_{t,t+1} R_{t+1}^l \left(1 - F_t\left(\bar{\xi}_{t+1}\right)\right) = \left(1 + \epsilon_{\eta_t}\right) - \epsilon_{\eta_t} \frac{\left(1 - \lambda_l\right) \eta_{t-1}}{\eta_t} \frac{1}{\pi_t} \frac{\left(Q_t^k - Q_t^l \eta_t^l\right)}{\Psi\left(\omega_t\right) \mu_t}, \tag{37}$$

$$K_t = \frac{\omega_t K_{t-1} \left(\tilde{\xi}_t R_t^k Q_{t-1}^k - \frac{\left(c_l + (1-\lambda_l) Q_t^l\right)}{\pi_t} \eta_{t-1}\right) + T_t^e}{Q_t^k - Q_t^l \eta_t}, \tag{38}$$

where equations (36) and (37) are the optimality conditions for equity injections

[17] All other parameters are set to the values described in Section IV.

[18] In our flexible price economy we implement this shock by removing the monetary policy rule and replacing it with $\pi_t = \exp(\epsilon_t^\pi)$ where ϵ_t^π is an i.i.d. random variable. As shown in Figures E.1 and E.2 in the Appendix, very similar results are obtained if we consider a one-time shock to the value of the outstanding debt, as captured by R_t^l, in our baseline model with nominal rigidities and $\nu = 0$.

and leverage in equations (12) and (17) repectively, using that $\Psi'(\omega_t) = 1 = \varphi_t = \gamma_t$ when $\psi = 0$. Equation (38) is the aggregate flow of funds constraint for entrepreneurs, equation (20), using equation (19) to substitute for aggregate equity.[19]

The level of capital demand is determined by equation (36), which requires that the rate of return on capital leaves the entrepreneurs indifferent between paying out dividends and injecting equity. This rate of return will depend on the leverage choice of entrepreneurs, which affects both the total amount of capital per unit of equity, $\frac{Q_t^k}{Q_t^k - Q_t^l \eta_t}$, and the expected default threshold $\bar{\xi}_{t+1}$. Leverage is determined by equation (37), where the last term captures the debt dilution channel associated with long-term debt. Without equity adjustment costs, capital demand is independent of net worth. Entrepreneurs are free to adjust equity injections ω_t to satisfy any given level of capital demand according to (38).

Figure 2 shows the response of the economy to an exogenous one-time decrease in inflation. The blue solid line shows the results of an economy with a debt duration of 6.5 years, that is $\lambda_l = 0.03$. The dotted red line shows results for an economy with short-term debt, i.e., $\lambda_l = 1$.

In both cases, the decline in inflation causes the real value of entrepreneurs' debt to increase and hence net worth to decline. In the economy with short-term debt, entrepreneurs react to the decline in net worth by reducing net dividend payouts, thus keeping investment, equity, and leverage unaffected. With short-term debt, the choice of leverage is not affected by the change in inflation (*cf.* the first-order condition in (37) with $\lambda_l = 1$). In contrast, with long-term debt, the increase in the real value of debt gives entrepreneurs an extra incentive to increase leverage, as captured by the last term in equation (37). The larger value of debt makes it more profitable for the entrepreneur to dilute the preexisting creditors by increasing leverage and hence decreasing the price of debt. The increase in leverage causes the expected default rate to go up and the required expected return on capital implied by optimal equity injections in equation (36) to rise. This increase in the return to capital required by equity investors in the economy with long-term debt captures a *debt overhang channel*, and causes investment to decline in response to a temporary decline in inflation.

B. Model with infinite equity adjustment costs

With $\psi = \infty$, the optimality conditions are given by:

[19] In the Appendix we show that when ψ is exactly equal to zero, the first order conditions of the entrepreneur's problem do not select an optimum but a saddle point. Therefore, we can think of the case with no equity adjustment cost as a calibration in which adjustment costs are arbitrarily small. In the Appendix we also show that as ψ goes to zero the derivative of the leverage policy function, η_η, goes to zero as well. See also Ajello, Perez-Orive and Szőke (2023) for additional analysis on the model with no equity adjustment costs.

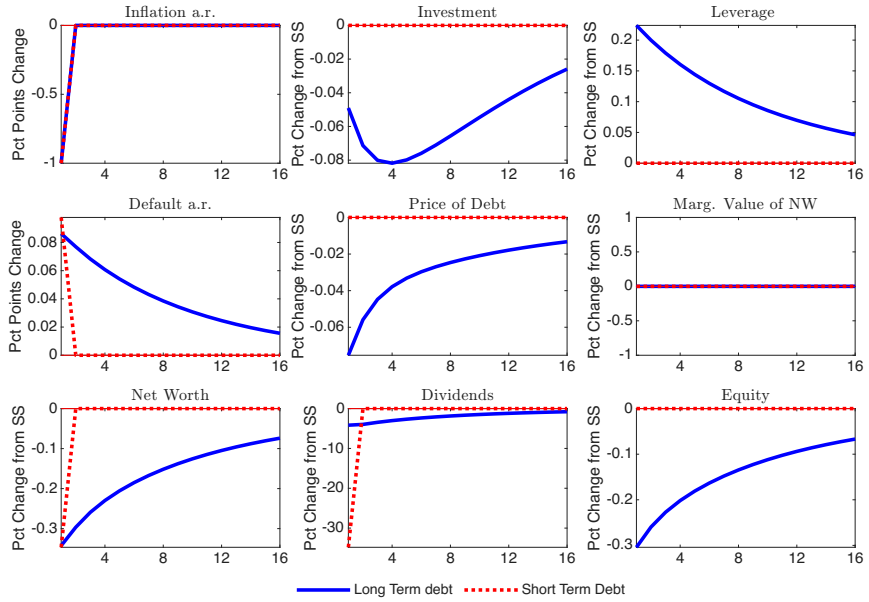


FIGURE 2. INFLATION SHOCK IN MODEL WITH FLEXIBLE PRICES: NO EQUITY ADJUSTMENT COSTS

Note: Figure shows the impulse response to a one time shock to inflation in a simple version of the main model with flexible prices and no real default costs. Inflation and default are shown as annual rates. Marg. Value of NW stands for 'marginal value of net worth', φ_t.

$$\omega_t = \bar{\omega}, \tag{39}$$

(40)

$$\mathbb{E}_t \Lambda_{t,t+1} R^l_{t+1} \left(1 - F_t\left(\bar{\xi}_{t+1}\right)\right) \varphi_{t+1} = \left(1 + \epsilon_{\eta_t}\right) \gamma_t - \epsilon_{\eta_t} \frac{\left(1-\lambda_l\right) \eta_{t-1}}{\eta_t} \frac{\left(Q^k_t - Q^l_t \eta^l_t\right)}{\Psi\left(\omega_t\right) \mu_t} \frac{\varphi_t}{\pi_t},$$

$$K_t = \frac{\bar{\omega} K_{t-1} \left(\tilde{\xi}_t R^k_t Q^k_{t-1} - \frac{\left(c_l + (1-\lambda_l) Q^l_t\right)}{\pi_t} \eta_{t-1} \right) + T^e_t}{Q^k_t - Q^l_t \eta_t}. \tag{41}$$

In this case, entrepreneurs cannot adjust equity injections, as shown in (39), so there is no active debt overhang channel in this economy.[20] Two features will determine the response of aggregate investment and output: the magnitude of the decline in net worth and the degree to which entrepreneurs increase leverage

[20] We set $\bar{\psi} = 0$ for simplicity.

in response to the shock.

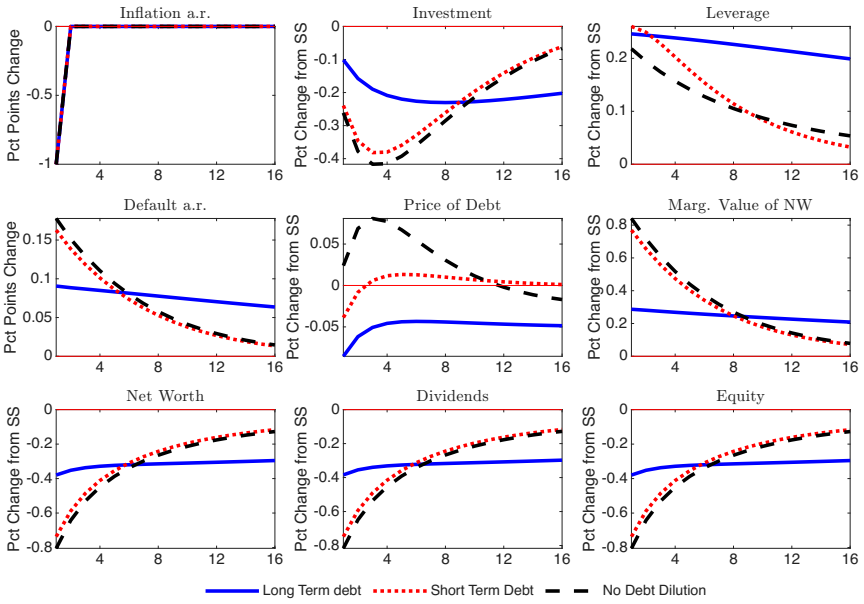


FIGURE 3. INFLATION SHOCK IN MODEL WITH FLEXIBLE PRICES: INFINITE EQUITY ADJUSTMENT COSTS

Note: Figure shows the impulse response to a one time shock to inflation in a simple version of the main model with flexible prices and no real default costs. Inflation and default are shown as annual rates. Marg. Value of NW stands for 'marginal value of net worth', φ_t.

Figure 3 shows the response of the economy with infinite equity adjustment costs to the same decrease in inflation used in Figure 2. The drop in net worth, caused by the increase in the real value of debt in equation (41), triggers the financial accelerator. Entrepreneurs borrow more to counter the decline in equity, but their ability to increase leverage is limited by the associated increase in the expected default rate. As a consequence, investment declines, further depressing net worth through a lower Q_t^k. This mechanism leads to a decline in investment both when debt is long-term (blue solid lines) and when debt is short-term (red dotted lines).[21] Importantly, however, the reduction in investment is now *dampened* in the economy with long-term debt. Two channels explain this result. First, through a *balance sheet channel*, the increase in leverage lowers the price of debt Q_t^l, which, from (41), dampens the fall in net worth. In Figure 3, net worth only falls by about half as much when debt is long-term. Second, because of *debt dilution*, represented by the last term in (40), the elasticity of leverage to a given

[21]The economy with infinite equity adjustment costs and short-term debt is very similar to the framework considered in Bernanke, Gertler and Gilchrist (1999) and Christiano, Motto and Rostagno (2014).

drop in net worth is larger with long-term debt. Equity declines half has much on impact in the economy with long term debt, but leverage increases by the same amount and then remains persistently elevated. Taken together, these two channels result in a decline in investment that is about fifty percent smaller.[22]

An interesting result in our model is that the same debt dilution incentive which, through debt overhang, is responsible for the decline in investment when there are no equity adjustment costs, actually causes a smaller contraction when it is impossible to adjust equity. To highlight this point, we included the purple dashed line in Figure 3, which presents the model responses when we shut down the debt dilution channel.[23] In this case, the impulse response functions are very close to those from the short-term debt model, suggesting that in this experiment the debt dilution effect is responsible for most of the dampening effects of a deflationary shock.[24]

The previous experiments discussed the effects of a stylized shock to inflation. As we discuss below, the dampening influence of long-term debt when equity adjustment costs are large extends to a wide range of shocks, especially financial shocks that affect leverage choices and balance-sheet dynamics. Furthermore, in our framework, the interaction between debt maturity and the behavior of financial variables is important for identifying the drivers of business cycle fluctuations, which we turn to next.

IV. Quantitative results

We now present the results of a quantitative version of our model. We briefly discuss the solution method, describe the parameterization and estimation of the model, and present the main insight of the paper, namely how the incorporation of long-term debt and equity adjustment costs affects the response of the economy to shocks.

A. Solution method

Defaultable long-term debt complicates the solution of the model by requiring the computation of the derivative of entrepreneurs' individual leverage policy function, $\eta_{\eta,t}(\eta_{t-1})$, which appears in (26). To overcome the associated computational challenge, we use an algorithm combining global methods for the solution

[22] The specific shock considered in Figure 3 causes both an increase in excess returns and in default rates. These two effects imply a sluggish recovery of entrepreneurial equity, generating a very persistent aggregate effect of the shock. The investment response has a half-life of 32 quarters. The very persistent response of investment is a property specific to the model and shock analyzed in Figure 3. As shown below, in our baseline quantitative model the investment response is much less persistent.

[23] In particular, we solve the model assuming that the last term in equation (40) is equal to zero. We can think of this alternative specification as a framework in which the outstanding debt is pooled across entrepreneurs, so that an individual entrepreneur does not internalize how his leverage decision affects the value of outstanding liabilities.

[24] In this exercise the movement in Q_t^l is fairly small, resulting in a small balance sheet effect. As we discuss below, this channel can be much stronger when an exogenous shock, like a monetary policy shock or a credit supply shock, directly lowers the price of outstanding debt.

of the model's steady state with perturbation methods to obtain impulse response functions.[25] In the first step of the algorithm, we use global methods to compute the policy function for leverage as a function of η_{t-1} and of a set of aggregate prices and quantities in steady state Γ^{SS}, that is, $\eta_t(\eta_{t-1}, \Gamma^{SS})$ and its derivative $\eta_{\eta,t} = \frac{\partial \eta_t}{\partial \eta_{t-1}} = g(\eta_{t-1}, \Gamma^{SS})$ in steady state. This result is achieved by an iterative procedure which solves for the model's steady state and for $\eta_{\eta,ss}$ simultaneously. Figure 4 shows the leverage policy function and the price of long-term debt on a grid of values for past individual leverage at the steady state of our model, as obtained from the first step of our algorithm. Leverage is upward sloping, due to the debt dilution incentive, and the derivative of this policy function at the steady state value of η is about 0.9. The right panel shows that Q^l is decreasing in leverage, as higher debt increases the probability of default.[26]

In the second step of the algorithm, we approximate the derivative function $g(\eta_{t-1}, \Gamma_t)$ with a linear function $\hat{g}(\eta_{t-1}, \Gamma_t)$ in the neighborhood of the steady state. This function can then be used as an additional equation of the model. As a result, standard perturbation techniques can be used to simulate the model dynamics. In addition, once we have calibrated the parameters determining the steady state values of entrepreneurial variables (such as equity adjustment costs, leverage, default rates, and spreads), this procedure allows us to use standard Bayesian techniques to estimate the model, as we discuss below. Section C.1 in the Appendix shows that our linear approximation of the derivative function is accurate.

B. Parameters

We divide the model parameters into three groups. The first group is calibrated to steady state moments, the equity adjustment cost parameter ψ is estimated using firm-level balance sheet data, and the remaining parameters are estimated using Bayesian methods. This approach guarantees that the approximation of the derivative of the entrepreneur's policy function, $\hat{g}(\hat{\eta}_{t-1}, \hat{\Gamma}_t)$, which is computed using global methods, is not affected by the values of the third group of parameters, making estimation feasible.

Calibrated parameters. — Table 1 reports the calibrated parameters, which are mostly set to standard values from the literature. For household preferences, we assume log utility and a discount factor that implies a steady state real interest rate of 2 percent. We choose an elasticity of substitution across goods- and labor varieties of six. Government spending is set to 20 percent of output in the steady state.[27]

[25] A similar algorithm was also used in Ferrante (2019). The Appendix describes the method in detail.

[26] In addition, we check numerically that the value function of the entrepreneur is concave in η in the neighborhood of the steady state.

[27] To account for the downward trend in inflation in the early part of the sample, we follow Christiano, Motto and Rostagno (2014) and calibrate the parameters governing the inflation target shock to $\rho_{\bar{\pi}} =$

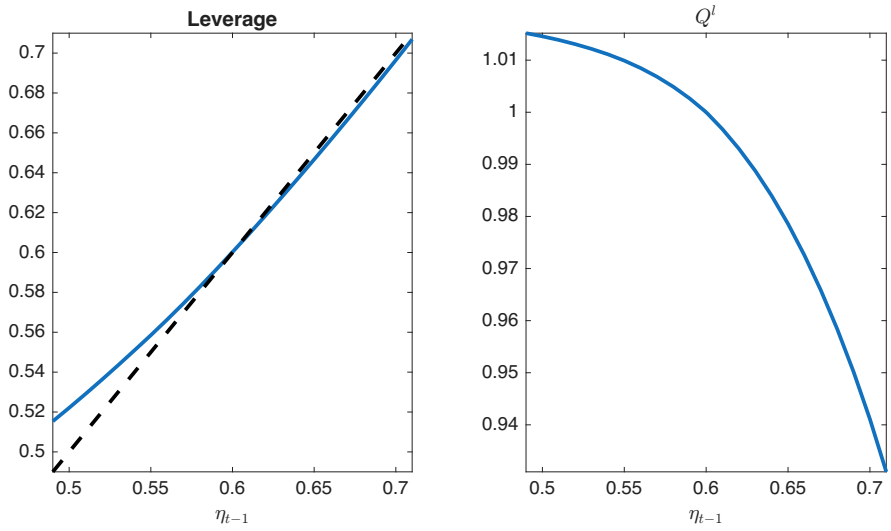


FIGURE 4. POLICY FUNCTIONS FOR THE INDIVIDUAL ENTREPRENEUR

Note: The left and right panels report the individual leverage policy function and the price of debt, Q^l, computed over a grid of past leverage around the steady state of the model. The black line in the left panel represents the 45 degree line.

The parameters pertaining to entrepreneurs are specific to our model. We set λ_l to obtain a debt duration of six and a half years, a value in line with the mean bond duration reported by Gilchrist and Zakrajšek (2012). The target dividend payout ratio is set to $1-\bar{\omega}=.01$ to obtain a steady state dividend/output ratio of around four percent, in line with the data. The parameters τ^e, σ_l, and γ_l are determined jointly. The targets are *i)* a leverage ratio of 0.6, *ii)* an annual default rate of three percent, *iii)* a bonds spread of 100 basis points annualized. The target for leverage implies a ratio of capital to net worth $K/N=2.5$ in line with the evidence in Kalemli-Ozcan, Sorensen and Yesiltas (2012). The default rate is the same used in Bernanke, Gertler and Gilchrist (1999) and in Ottonello and Winberry (2020). The spread is calibrated to the average value of the difference between Moody's BAA corporate yield and the AAA corporate yield between 1986 and 2019. Finally we normalize $\bar{\psi}$ so that in steady state there are no quadratic costs of issuing equity, i.e. $\omega_{ss}=\bar{\omega}$.[28]

ESTIMATION OF EQUITY ADJUSTMENT COSTS. — The last row of Table 1 shows our estimated value of ψ, the parameter governing the quadratic equity adjustment costs. To a first order, this parameter only affects the optimality condition for

0.975 and $\sigma_{\bar{\pi}}=0.0001$.

[28] We assume that default costs entail a resource loss ($\nu=1$) as is standard in the literature.

entrepreneurs' equity injections. As we described in Section III, this parameter plays a key role in our model because it determines how long-term debt affects the response of the economy to shocks. To estimate ψ we use the optimality condition for equity injections, (9), which we reprint here for convenience:

$$1 = \frac{\Psi'(\omega_t)}{Q_t^k - Q_t^l \eta_t} \cdot \mathbb{E}_t \Lambda_{t,t+1} \int_{\bar{\xi}_{t+1}}^{\infty} v_{t+1}(\eta_t, \xi_{t+1})\, dF_t(\xi_{t+1})$$

We define the return on equity injections as

$$R_{t+1}^x = \frac{\int_{\bar{\xi}_{t+1}}^{\infty} v_{t+1}(\eta_t, \xi_{t+1})\, dF_t(\xi_{t+1})}{Q_t^k - Q_t^l \eta_t}$$

and use the functional form for equity adjustment costs, $\Psi'(\omega) = 1 - \bar{\psi} - \psi(\omega_t - \bar{\omega})$ to write

$$\mathbb{E}_t \Lambda_{t,t+1} R_{t+1}^x = \frac{1}{1 - \bar{\psi} - \psi(\omega_t - \bar{\omega})}, \tag{42}$$

which we linearize to obtain

$$\mathbb{E}_t r_{t+1}^x - r_{t+1} = \frac{\psi}{1 - \bar{\psi}} d\omega_t. \tag{43}$$

Here r_{t+1}^x and r_{t+1} denote percent deviation of R_{t+1}^x and the real risk-free rate from their steady state values. We then use firm-level balance sheet data from Compustat (S&P Global Market Intelligence, 2025) to estimate this relationship in the data.

Our sample consists of 19,341 U.S. non-financial firms between 1985q1 and 2023q1.[29] In the data, we define the return on equity as firm i's time $t+1$ cum-dividend market value of equity divided by its time t cum-dividend book value of equity, i.e., $R_{i,t+1}^x = \frac{V_{i,t+1}}{X_{i,t}}$. The end-of-quarter market value of equity, $V_{i,t+1}$, is measured using Compustat items `cshoq` and `prccq`. The book value of equity, $X_{i,t}$, is given by `ceqq`. Net dividend payouts are given by dividends (`dvy`), minus net repurchases (`sstkyq - prstkcyq`). This lets us define ω_{it} as one minus the fraction of net payouts to shareholders over the cum-dividend book value of equity. We run the following regression:

$$\frac{R_{i,t+1}^x}{1 + r_{t+1}} = \psi \omega_{i,t} + \delta_i + \delta_t + \nu_{it+1} \tag{44}$$

Here r_{t+1} is the quarterly realized real rate and δ_i and δ_t are firm- and time fixed

[29] Details are presented in Appendix D.

effects.[30]

This approach produces a value of $\psi = 10.9$ with a standard error of 0.13. The result is robust to the inclusion of the fixed effects. Our estimate of ψ implies significant equity adjustment costs, resulting in model dynamics close to those of a calibration with $\psi = \infty$.[31]

Estimated parameters. — We estimate the remaining parameters of the model using Bayesian methods. We use quarterly observations on ten variables between 1985Q4 and 2019Q4. Those include seven standard macroeconomic variables (GDP, investment, consumption, hours worked, inflation, real wages, and the Federal Funds Rate) in addition to three financial variables. The latter include the spread between BAA-rated corporate bonds and the ten-year US government bond rate, total credit to non-financial firms from the U.S. Financial Accounts (Board of Governors of the Federal Reserve System, 2026*a*), and equity valuations from the Wilshire 5000 index. The model equivalents to the financial variables are the long-term debt spread, $\mathbb{E}_t \frac{\sum_{i=1}^{28}(R^l_{t+i} - R^d_{t+i})}{7}$, total credit $Q^l_t L_t$, and total equity X_t. The underlying series are obtained from Haver Analytics (Haver Analytics, 2026); the hourly-compensation, BAA-yield, and ten-year Treasury series are obtained from FRED (U.S. Bureau of Labor Statistics, 2026; Moody's, 2026; Board of Governors of the Federal Reserve System, 2026*b*); and the shadow Federal Funds Rate is from the Wu–Xia data set (Wu and Xia, 2026). We take log differences for GDP, investment, consumption, credit, equity, and real wages, and we remove the sample mean from all observable variables.[32]

Table 2 reports the priors and posteriors for the estimated parameters. The priors are calibrated as in Christiano, Motto and Rostagno (2014). The posterior standard deviations are for the most part significantly smaller than the standard deviation of the prior distributions, suggesting that the data are informative about the estimated parameters. The top panel of Table 2 reports the values for the economic parameters, which are within the range of estimates in the literature. The second panel reports the results for the shock autocorrelation parameters. Most of the shocks are quite persistent, with the exception of the wage markup shock. The standard deviations of the macroeconomic shocks are reported in the bottom panel of Table 2. One thing to notice is that our estimation implies that

[30] We omit $\bar{\psi}$ which, being a very small number, does not affect the results in any meaningful way.

[31] Appendix D presents additional details on the robustness of our estimation. Appendix E compares the impulse responses of our baseline economy to those from a calibration with $\psi = \infty$.

[32] Real consumption is the sum of nondurable goods and services. Real investment is obtained as the sum of gross private domestic investment plus durable goods purchases. Aggregate hours are an index of nonfarm business hours for all persons. These variables are converted into real, per-capita, terms by dividing them by the population over 16 and by the GDP implicit price deflator. Inflation is measured as the change in the GDP price deflator. Real wages are given by the hourly compensation of all nonfarm employees divided by the GDP deflator. For the Federal Funds Rate, we use the shadow rate developed by Wu and Xia (2016). Credit is measured as total liabilities for nonfarm, non-financial corporate business from the Financial Accounts. As equity, measured by the Wilshire 5000 index, it is converted to real per-capita terms as above.

the standard deviation of the anticipated component of the risk shock $\sigma_{\sigma,n}$ is larger than the one on the unanticipated component σ_σ. As a result, most of the fluctuations in σ_t are due to anticipated shocks.

C. *Credit supply shocks and business cycle fluctuations*

The main takeaway from our estimated model with long-term debt and costly equity issuance is that the credit supply shock is the main driver of business cycle fluctuations. We present this result in Table 3, which reports the unconditional variance decomposition of key macroeconomic and financial variables. The table is divided into four panels. The first panel reports the variance decomposition for our estimated baseline model, which includes long-term debt and costly equity issuance. It shows that the credit supply shock is the main driver of output fluctuations, accounting for roughly half of the business cycle variance of GDP growth and investment growth. The credit supply shock also explains a large part of the fluctuations in financial variables, accounting for 45 percent of the variance of the Federal Funds Rate, 20 percent of the variance of credit growth and 17 percent of the variance of equity fluctuations.[33]

Figure 5 complements the results in the first panel of Table 3 by conducting an in-sample exercise that compares the behavior of real and financial variables from the data (blue solid line) with a model simulation, obtained by feeding only the estimated credit supply shocks into the model (black starred line). The top left panel shows that the credit supply shock does well at reproducing the time series for output growth over our estimation sample. In particular, the large contraction in output during the 2008 financial crisis is associated with a large negative credit supply shock. The credit supply shock also tracks the behavior of hours and the policy rate well. The remaining panels in Figure 5 show that the shock generates financial time series in line with the data, especially for credit growth and credit spreads.

Finally, we show that our estimate for the equity adjustment cost parameter ψ, obtained from cross-sectional data, is consistent with the time series behavior of dividends in the data. The dividend-to-GDP ratio implied by the model, which is compared to the data in Figure 6, is obtained by feeding the estimated credit supply shocks into the model. Although this variable was not used in the estimation, the model produces a time series that is close to its empirical counterpart. In particular, the model captures the procyclical nature of dividend payouts. This evidence suggests that our estimate for ψ results in a realistic time series behavior for aggregate dividends in the model.

The principal reason why the credit supply shock is identified as the driving force of the business cycle in our model is that it delivers impulse responses with

[33]The credit supply shock accounts for a small share of the unconditional variance of spreads, while the bulk of this variable is explained by the risk shock. This result is likely due to the fact that the additional news component of the risk shock, together with its higher persistence, allows this shock to affect the forward-looking spread variable in our model more directly.

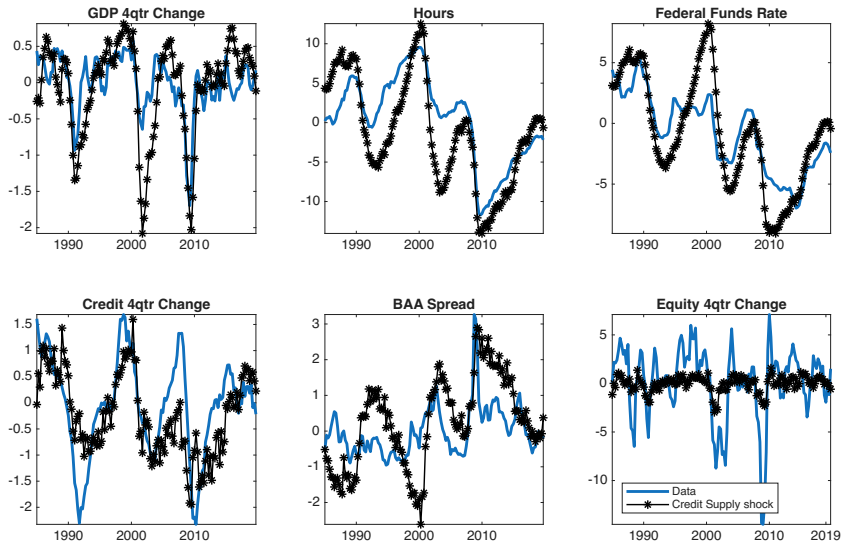


FIGURE 5. THE ROLE OF THE CREDIT SUPPLY SHOCK

Note: Figure compares the behavior of selected variables in the data (purple lines) with the model-implied behavior attributed to the filtered credit supply shocks (blue lines). Variables are reported in deviation from their mean. For GDP, credit and equity we report the four quarter change.

the right co-movement between real and financial variables. Figure 7 illustrates this result by reporting the impulse responses to a credit supply shock, shown as the blue solid lines. For comparison, we also include two other shocks the literature has identified as important drivers of economic activity: a risk shock (black dashed line) and a shock to the marginal efficiency of investment (red dotted line). The credit supply shock reduces lenders' appetite for corporate debt and increases credit spreads by about 40 basis points. With sizeable equity issuance costs, entrepreneurs do not make up for the decline in debt with external equity. As a result, entrepreneurs cut investment, putting downward pressure on asset prices and causing equity to decline. This mechanism implies a standard financial accelerator channel as in Bernanke, Gertler and Gilchrist (1999), which results in a large contraction in investment and output. Consumption initially rises in response to lower interest rates, but eventually declines persistently as lower wages drag down demand.[34] The credit supply shock delivers a co-movement between real and financial variables consistent with the data.

[34]The response of consumption to a credit supply shock depends on how sticky prices are and on how aggressive monetary policy is in responding to inflation and output. Figure E.3 in the appendix shows that with stickier prices and less aggressive monetary policy, consumption would decline on impact in response to a credit supply shock.

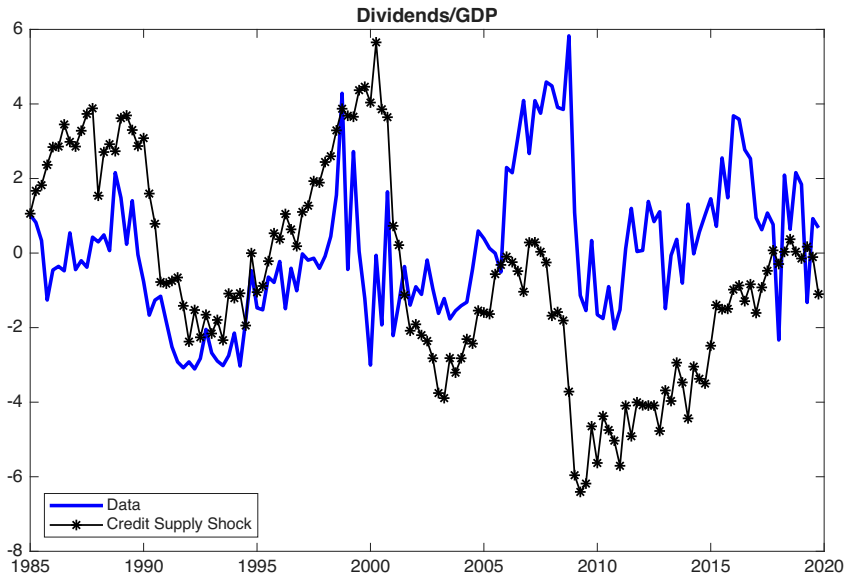


FIGURE 6. THE ROLE OF THE CREDIT SUPPLY SHOCK: DIVIDENDS/GDP

Note: Figure compares the behavior of Dividends/GDP in the data (purple lines) with the model-implied behavior attributed to the filtered credit supply shocks (blue lines). Variables are reported in deviation from their mean. Dividends are measured as in Jermann and Quadrini (2012).

In contrast, as shown in the bottom row of Figure 7, the risk shock and the shock to the marginal efficiency of investment, both induce a countercyclical movement of equity, which is at odds with the data. The shock to the marginal efficiency of investment is an exogenous contraction in the supply of investment goods that causes investment and output to decline and the price of capital to increase. As a result, entrepreneurs' asset values - and with it equity - rise during the downturn. This counterfactual behavior of equity, following a shock to the marginal efficiency of investment, is discussed by Christiano, Motto and Rostagno (2014) as the main reason in favor of risk shocks in a model with short-term debt. With long-term debt, however, the risk shock *also* leads to a counter-cyclical response of equity. When $\lambda_l < 1$, the risk shock causes a large increase in entrepreneurs' debt dilution incentive by making the price of debt, Q_t^l, more sensitive to leverage. This is because with a larger variance of idiosyncratic shocks, the density around the default threshold increases and hence an increase in leverage causes a larger increase in default and a larger drop in the price of debt, as suggested by equations (24) and (26). Figure 8 illustrates this point by showing the responses of the debt dilution incentive (the last term in (17)), together with the prices of debt and assets after a credit supply shock (blue solid line) and a risk shock (black dashed

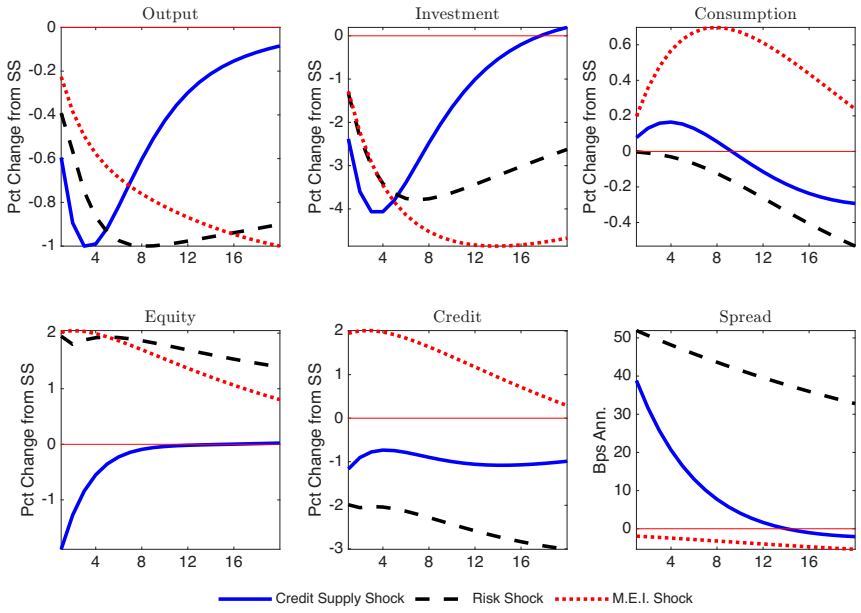


FIGURE 7. CREDIT SUPPLY SHOCK, RISK SHOCK, AND MEI SHOCK

Note: Figure shows the impulse response to a credit supply shock (blue solid line), a risk shock (black dashed line), and a shock to the marginal efficiency of investment (red dotted line) in our estimated model.

line). The large increase in the debt dilution incentive following a risk shock causes a larger drop in the value of debt and a smaller drop in the price of capital compared to the responses to a credit supply shock. As a result, equity increases after a risk shock.[35]

To highlight the importance of including equity prices among the variables used in the estimation, the second panel of Table 3 shows that if we remove equity as an observable, shocks to the marginal efficiency of investment become the main driver business cycles as in Justiniano, Primiceri and Tambalotti (2010).

THE ROLE OF LONG-TERM DEBT. — Assumptions on corporate debt maturity have important implications for the inference of which shocks drive the business cycle. Restricting the model to short-term debt shifts the balance from credit supply shocks to risk shocks, as we show in the third panel of Table 3. This panel reports the variance decomposition obtained when re-estimating our model under the assumption that debt is short-term, i.e., $\lambda_l = 1$. In this case, the risk shock

[35]Figure E.4 in the Appendix shows the impulse response to a net worth shock in our model. As suggested by Christiano, Motto and Rostagno (2014), this shock generates a countercyclical response of credit.

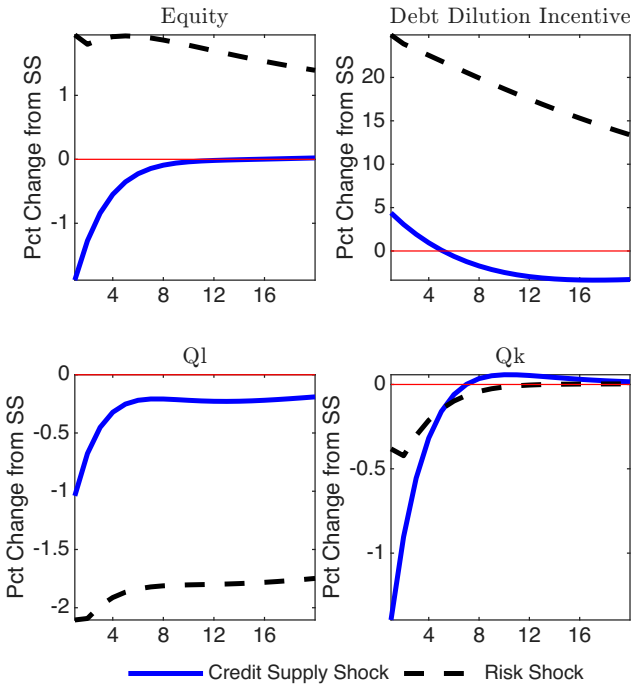


FIGURE 8. EQUITY RESPONSE TO CREDIT SHOCK AND RISK SHOCK

Note: Figure reports the impulse response to a credit supply shock (blue line) and to a risk shock (green dotted line) in our estimated model. The debt dilution incentive variable is the last term in equation (17).

becomes the main driver of fluctuations in output and investment. This result is in line with Christiano, Motto and Rostagno (2014) who find that, in a model with short-term debt, fluctuations in entrepreneurial risk, σ_t, are the most important shock for the business cycle.[36]

The reason for this result is that in a model with short-term debt an increase in risk induces the right co-movement between macro and financial variables. This point is illustrated in Figure 9, which shows the economy's response to a risk shock in our baseline (blue solid line) and in the estimated model with short-term debt (red dotted line). The risk shock increases the dispersion of future idiosyncratic

[36]When setting $\lambda_l = 1$, our model is very similar to that used in Christiano, Motto and Rostagno (2014), other than the following key differences in the modeling assumptions and the estimation strategy: *i)* Christiano, Motto and Rostagno (2014) use the contractual framework of Bernanke, Gertler and Gilchrist (1999) which assumes that entrepreneurs bear all the risk from aggregate shocks, whereas we let lenders face credit risk; *ii)* we use a longer estimation sample and a slightly different set of shocks; *iii)* in the estimation, Christiano, Motto and Rostagno (2014) match the BAA spread, a measure based on long-term securities, with the realized one-period spread in their model; *iv)* in our baseline calibration $\psi = 10.86$, whereas Christiano, Motto and Rostagno (2014), following Bernanke, Gertler and Gilchrist (1999), assume $\psi = \infty$.

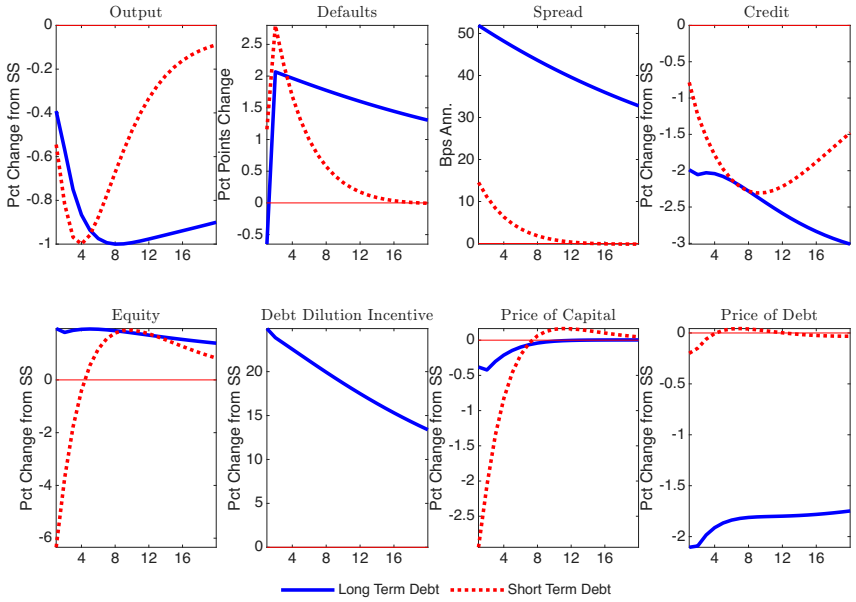


FIGURE 9. RISK SHOCK IN BASELINE MODEL AND IN MODEL WITH SHORT-TERM DEBT

Note: Figure reports the impulse responses to a risk shock in our estimated baseline model with long-term debt (blue line) and in the estimated model with short-term debt (red dashed line). Shocks are normalized to deliver a one percent decline in output.

entrepreneurial shocks, implying persistently higher default rates and causing spreads to rise and credit to decline. When debt is short-term, the debt dilution incentive is absent and there is no balance sheet effect from fluctuations in the price of debt. As a result, an increase in risk causes a larger drop in asset prices than in the case with long-term debt, implying a decline in entrepreneurs' equity. The risk shock thus induces a pro-cyclical response of equity prices in a model with short-term debt, in line with the data.

THE ROLE OF EQUITY ADJUSTMENT COSTS. — The second key ingredient of our estimated model are the equity adjustment costs. The bottom panel of Table 3 shows that if we estimate the model with $\psi = 0$, we obtain a much smaller role for the credit supply shock, whereas the risk shock is again the main driver of fluctuations in output and investment.[37] Figure 10 shows the impulse response to a credit shock (blue solid line) and a risk shock (black dashed line) in the estimated model without equity adjustment costs. Notice that the exogenous increase in the interest rate on entrepreneurs' liabilities, caused by the credit supply shock,

[37]When estimating the model with $\psi = 0$, we keep the same steady state targets from Table 1, but adjust the bond spread to obtain a similar value of σ_l.

now results in an *increase* in output. When equity adjustments are costless, entrepreneurs respond to a credit supply shock by substituting debt issuance with equity issuance. As a result, leverage and default rates fall, weakening the debt overhang effect. This leads to investment and output increasing, while credit spreads tighten and total debt declines. This counterfactual countercyclical behavior of debt in response to credit supply shocks leads the estimation to favor the risk shock in this case. However, the marginal data density of the model with $\psi = 0$ is 4684, more than 300 points below the marginal data density in our baseline model of 4998.

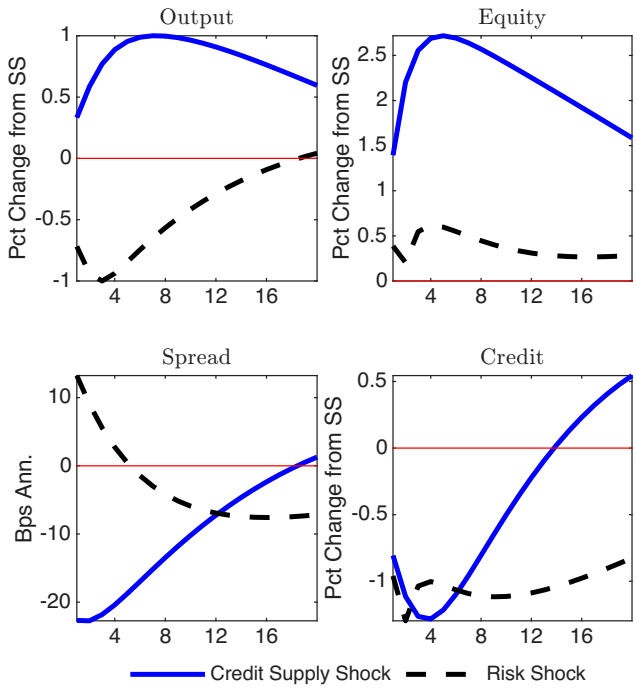


FIGURE 10. CREDIT SHOCK AND RISK SHOCK IN MODEL WITHOUT EQUITY ADJUSTMENT COSTS

Note: Figure reports the impulse response to a credit supply shock (blue line) and to a risk shock (green dotted line) in a model with no equity adjustment costs ($\psi = 0$).

While without equity adjustment costs the model dynamics change substantially, we show that our baseline calibration behaves similarly to a model with infinite adjustment costs. Figure E.5 in the Appendix shows that the impulse responses in our baseline model and in a model with $\psi = \infty$ are very close. In addition, Table E.1 reports the variance decomposition when we re-estimate the model with $\psi = \infty$ and shows that the results are similar to those obtained in

our baseline calibration with $\psi = 10.86$ (top panel of Table 3).[38] These results suggest that our estimated value for ψ results in substantial equity frictions.

Overall, the results in Table 3 suggest that to capture the importance of credit supply shocks in an estimated DSGE model with financial frictions, three elements are necessary: *i)* using financial variables as observables, *ii)* the presence of long-term debt, and *iii)* having equity adjustment costs consistent with the data.

D. Debt maturity and macroeconomic shocks

In Section III, we studied how corporate debt maturity affects the response to a stylized shock to inflation in a simplified version of our model. Figure 11 shows how debt maturity affects the transmission of standard macroeconomic shocks by comparing the impulse responses to five different shocks in the baseline model with long-term debt (blue line) and in a model with an identical parameterization but with short-term debt (red dashed line).

The top three panels of Figure 11 suggest that in a model with realistic equity adjustment costs, long-term debt generates an important attenuation in the transmission of financial shocks which affect investment demand, such as a credit supply shock, a risk shock, or a monetary policy shock. As discussed in Section III, this result is mainly due to the positive effect of a decline in the price of debt on net worth (balance sheet channel) and to the higher elasticity of leverage (debt dilution channel). Compared to a model with short-term debt, these channels result in a higher path for equity and in a decline in output that is between 30 to 70 percent smaller. The mitigating effect of long-term debt is similar for a credit supply shock and for a contractionary monetary policy shock. The latter result is consistent with the evidence in Jungherr et al. (2024), who show that investment by firms with a larger share of long term debt is less responsive to monetary policy shocks. For the risk shock, the dampening effect of long term debt is particularly strong, and it results in a counterfactual increase in equity which was key for the estimation for the baseline model, as we discussed in the previous section.[39]

The fourth row shows impulse responses to a shock to the marginal efficiency of investment. Unlike the other shocks, long-term debt amplifies the near-term response of output to a marginal efficiency of investment shock. This result is due to the behavior of the price of debt, which increases due to the decline in the real rate, implying a lower path of equity, and hence lower investment and output.

Finally, the last row shows the impulse response to a TFP shock. In this case debt maturity does not affect the behavior of output in a meaningful way, because equity and the price of debt move very little.

[38] However, the marginal data density with $\psi = \infty$ is 4993, lower than in our baseline specification.

[39] In Figure E.6 we repeat the exercise from Figure 11 while assuming that $\psi = \infty$. The gap between the response to financial shocks of the economy with long-term debt and the one with short-term debt becomes even larger, suggesting that the balance sheet channel and the debt dilution channel strengthen as ψ increases.

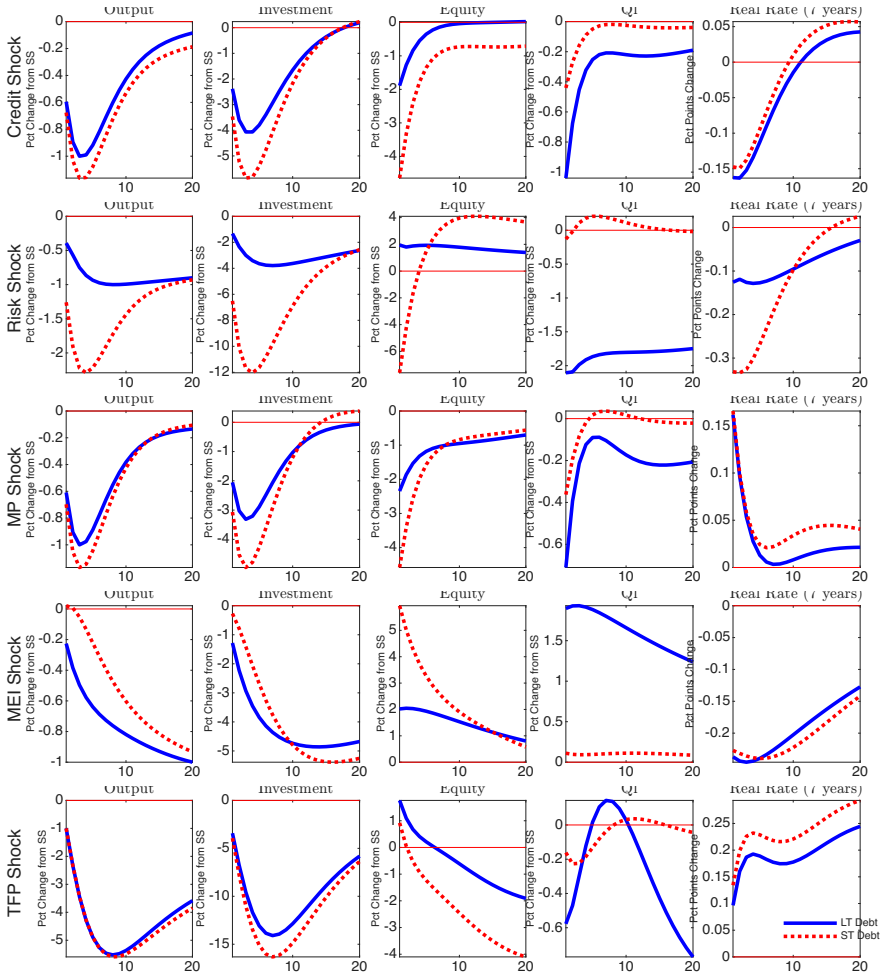


FIGURE 11. MACROECONOMIC SHOCKS: THE ROLE OF LONG-TERM DEBT

Note: Figure compares the impulse responses in our estimated baseline model (blue solid line) to a model with identical parameters except that all debt is short-term (red dashed line). The real rate in the figure is computed as $\mathbb{E}_t \frac{\sum_{i=0}^{27} R_{t+i}/\pi_{t+i+1}}{7}$. All shocks are rescaled to obtain a one percent drop in output in our baseline model with long-term debt.

V. A model with financially constrained intermediaries

In our baseline model, financial intermediaries raise deposits from households and hold entrepreneurial debt without facing any financial frictions. In this section, we extend the model to allow for financial constraints on intermediaries, following the framework of Gertler and Karadi (2011) and Ferrante (2019). The extended model serves two purposes. First, we show that when intermediaries are financially constrained, the effects of a negative shock to entrepreneurial net

worth continue to be dampened by the presence of long-term debt. This dampening can be decomposed into two components: a debt dilution channel and a channel operating through an endogenous decline in the price of entrepreneurial debt. The latter mechanism becomes quantitatively more important when banks have limited risk-bearing capacity, as a decline in bank net worth reduces their demand for loans. Second, while in the baseline model exogenous variation in banks' required rate of return is the primary source of business cycle fluctuations, the extended framework allows us to reinterpret these credit supply shocks as shocks to the net worth of financially constrained intermediaries.

The key equilibrium condition affected by the introduction of financial constraints on intermediaries is the demand for entrepreneurial debt. In the baseline model, this demand is characterized by equation (23), which we reproduce here for convenience:

$$\mathbb{E}_t \Lambda_{t,t+1} \kappa_t R^b_{t+1} = 1. \tag{45}$$

As we show in Appendix F, with financially constrained intermediaries, the pricing condition for entrepreneurial debt instead becomes

$$\mathbb{E}_t \Lambda^b_{t,t+1} R^b_{t+1} = 1. \tag{46}$$

where $\Lambda^b_{t,t+1}$ denotes the stochastic discount factor of financial intermediaries. It is given by

$$\Lambda^b_{t+1} = \frac{\Omega_{t+1}}{\theta^l + E_t \Omega_{t+1} \frac{R^d_t}{\pi_{t+1}} \left(\frac{\phi^b_t - 1}{\phi^b_t}\right)}. \tag{47}$$

Here, Ω_t measures the marginal value of intermediates' wealth, and ϕ^b_t denotes their leverage, defined as the ratio of assets to net worth, $\phi^b_t = \frac{Q^l_t L_t}{N^b_t}$. Low net worth raises leverage, which lowers the stochastic discount factor and the required rate of return on loans relative to the deposit rate. This mechanism is similar to the transmission of the exogenous credit supply shock that we consider in our baseline framework.

Figure 12 shows the economy's response to a negative shock to entrepreneurial net worth.[40] When entrepreneurial debt is long term (solid blue line), the contraction in entrepreneurial equity leads to a fall in the price of debt and a decline in intermediaries' net worth. Intermediaries respond by requiring a higher spread between loan and deposit rates, increasing the liquidity premium. Despite this larger endogenous increase in the liquidity premium, the model with long term debt exhibits a shallower economic contraction. The reason is that the larger decline in the price of debt cushions the drop in entrepreneurial net worth. As

[40] Appendix F provides additional details on the calibration of the extended model.

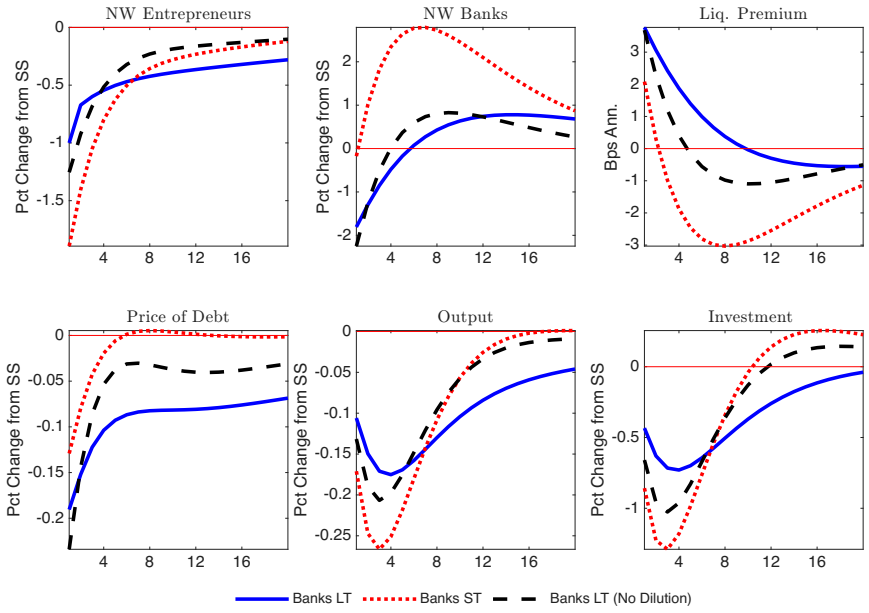


FIGURE 12. ENTREPRENEURIAL NET WORTH SHOCK IN MODEL WITH CONSTRAINED INTERMEDIARIES

Note: Figure reports the impulse responses to shock to entrepreneurial net worth in a model extension with constrained financial intermediaries. The blue line represents a model with long term debt, the black dashed line a model with long term debt but no debt dilution channel and the red dotted line represents a model with short term debt. The shock is calibrated to deliver a 1 percent decline in entrepreneurial net worth in the model with long term debt.

shown by the black dashed line, this balance sheet effect is strong enough to dampen the model response to an entrepreneurial net worth shock even when the debt dilution incentive is shut down.[41]

Figure 13 compares the response to a credit supply shock in the baseline model with the response to a shock to intermediaries' net worth in the economy with financially constrained intermediaries. The negative net worth shock to intermediaries generate an endogenous increase in liquidity premia that is almost identical to the exogenous increase in the baseline framework. As a result, the macroeconomic responses to the two shocks are nearly identical across the two economies. This result provides an alternative interpretation of the baseline credit supply shock as a reduced-form representation of intermediary balance sheet disturbances.

[41] Appendix F reports similar results for other financial shocks in our extended model.

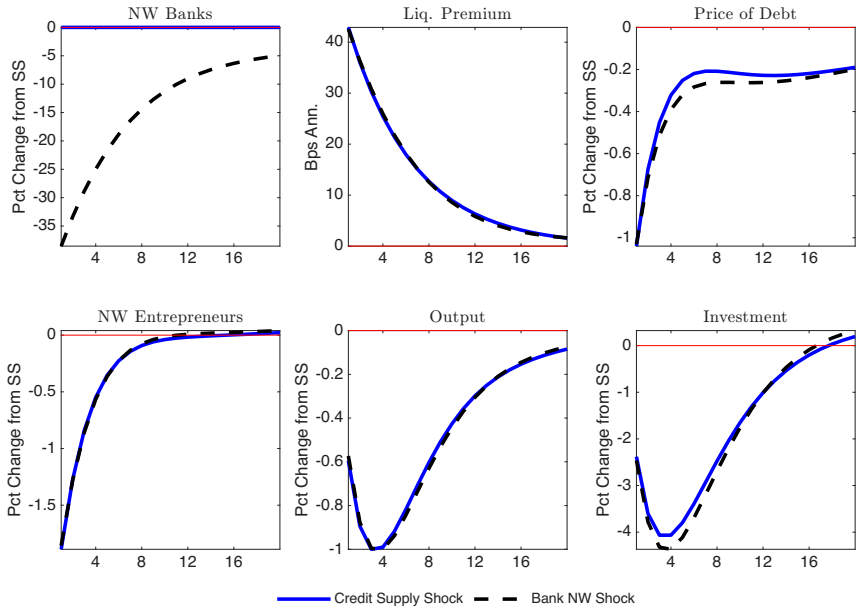


FIGURE 13. CREDIT SUPPLY SHOCK AND BANK NET WORTH SHOCK

Note: Figure reports the impulse response to a credit supply shock (blue line) in our baseline model and to a bank net worth shock (black dashed line) in a model extension with constrained financial intermediaries. Shocks are rescaled to generate the same decline in output.

VI. Conclusion

This paper develops and estimates a medium-scale New-Keynesian model with long-term corporate debt and costly equity issuance. We show that the interplay between long-term debt and equity adjustment costs fundamentally alters the transmission of macroeconomic shocks. When equity injections are costly, the debt dilution incentive and the resulting balance-sheet channel both work to partially offset declines in net worth following adverse shocks. This mechanism dampens the contraction of investment and output in response to financial shocks.

Incorporating realistic financing frictions into the model leads us to a re-assess the drivers of business cycle fluctuations. Our framework identifies a credit supply shock as the primary source of variability in output and investment, as this is the only shock that generates the observed co-movement between macroeconomic aggregates and financial variables. While in our baseline we capture the shock to credit supply in a reduced form way, in an extension with financially constrained intermediaries we show that the macroeconomic effects of our credit supply shock are nearly identical to the effects of a shock to the net worth of financial intermediaries.

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Table 1—Calibrated parameters

Parameter	Value	Description	Target/Source
Households			
β	0.995	Discount factor	2% Real rate
σ	1	IES	Standard
φ	1	Inverse Frisch El.	Standard
Intermediate Production			
χ	0.4	Capital share	Standard
ε_p	6	Ela. of subst. goods	Standard
ε_w	6	Ela. of subst. labor	Standard
Investment Production			
δ_k	0.025	Depreciation rate	Standard
Government			
G/Y	0.2	Gov. exp. to GDP	Standard
Entrepreneurs			
λ_l	0.0305	Debt amortization	Debt duration: 6.5 yrs.
$1-\bar{\omega}$	0.01	Dividend payout	4% Dividend/GDP
τ^e	0.01	Transfer new entrants	$\eta^l = 0.6$
σ_l	0.20	St. Dev. idiosyncratic risk	3% Default rate
γ_l	0.72	Recovery rate	$R^l - R = 100\,\text{bp}$
$\bar{\psi}$	0.0054	Linear equity iss. cost	$\omega_{ss} = \bar{\omega}$
ν	1	Resource cost of default	Literature
ψ	10.86	Quadratic equity iss. cost	Compustat

Table 2—Estimated parameters

			Prior		Posterior	
Parameter	**Description**	**Prior Dist.**	**Mean**	**S.D.**	**Mode**	**S.D.**
Economic parameters						
s_p	Slope price Phillips curve	Beta	0.05	0.01	0.0435	0.0068
s_w	Slope wage Phillips curve	Beta	0.05	0.01	0.0167	0.0033
ι_p	Price indexing	Beta	0.50	0.15	0.9017	0.0448
ι_w	Wage indexing	Beta	0.50	0.15	0.4657	0.1569
κ_π	Policy weight inflation	Normal	1.50	0.25	2.6103	0.1342
κ_y	Policy weight growth	Normal	0.25	0.10	0.6610	0.0760
ρ_i	Policy inertia	Beta	0.75	0.10	0.8432	0.0121
h	Habit	Beta	0.50	0.10	0.8346	0.0228
γ_I	Investment costs	Normal	5.00	3.00	1.1939	0.1353
γ_u	Utilization costs	Normal	2.00	1.00	4.7088	0.6900
Autocorrelation parameters						
ρ_κ	Credit supply	Beta	0.50	0.20	0.8403	0.0124
ρ_σ	Risk	Beta	0.50	0.20	0.9845	0.0050
$\rho_{\sigma,n}$	Risk news	Beta	0.50	0.20	0.6889	0.0669
ρ_I	M.E.I.	Beta	0.50	0.20	0.9954	0.0020
ρ_ε	Price markup	Beta	0.50	0.20	0.7994	0.0362
ρ_z	TFP	Beta	0.50	0.20	0.9602	0.0137
ρ_g	Government	Beta	0.50	0.20	0.9455	0.0152
ρ_c	Preference	Beta	0.50	0.20	0.8109	0.0413
ρ_{ε_w}	Wage markup	Beta	0.50	0.20	0.0993	0.0635
Standard deviation parameters						
σ_κ	Credit supply	IG2	0.002	0.0033	0.0053	0.0005
σ_σ	Risk	IG2	0.002	0.0033	0.0008	0.0004
$\sigma_{\sigma,n}$	Risk news	IG2	0.002	0.0033	0.0192	0.0016
σ_I	M.E.I.	IG2	0.002	0.0033	0.0208	0.0012
σ_ε	Price markup	IG2	0.002	0.0033	0.0009	0.0001
σ_z	TFP	IG2	0.002	0.0033	0.0049	0.0004
σ_g	Government	IG2	0.002	0.0033	0.0159	0.0010
σ_c	Preference	IG2	0.002	0.0033	0.0276	0.0036
σ_{ε_w}	Wage markup	IG2	0.002	0.0033	0.0091	0.0007
σ_N	Equity shock	IG2	0.002	0.0033	0.0338	0.0021
σ_R	Monetary policy	IG2	0.583	0.8250	0.4666	0.0343

Table 3—Variance decomposition

	Credit Supply	Risk	M.E.I.	NW	Markup	Gov.	M.P.	TFP
	Baseline model							
GDP	45	8	6	6	17	7	3	4
Investment	52	9	14	6	13	0	2	2
Consumption	7	6	23	1	2	1	0	4
Spread	8	78	12	2	0	0	0	0
Equity	17	2	9	69	1	0	1	0
Credit	20	11	33	31	1	1	1	0
Inflation	36	13	9	1	19	1	4	6
FFR	45	19	25	2	1	1	2	1
Hours	28	11	32	1	16	1	1	2
Wages	2	2	3	0	10	0	0	3
	Model with long-term debt – no equity in estimation							
GDP	7	2	27	2	17	17	5	6
Investment	30	3	42	4	12	0	3	3
Consumption	13	0	1	0	3	2	1	6
Spread	54	25	0	10	0	1	0	1
Equity	13	0	0	72	1	1	5	0
Credit	19	4	5	44	6	2	4	1
Inflation	17	2	5	1	34	2	8	6
FFR	40	3	9	2	4	3	7	2
Hours	20	2	10	1	23	5	3	4
Wages	1	0	0	0	10	0	0	3
	Model with short-term debt							
GDP	5	24	4	9	24	10	5	8
Investment	11	35	13	12	17	0	4	4
Consumption	6	4	15	1	3	1	0	7
Spread	99	1	0	0	0	0	0	0
Equity	5	10	29	48	3	0	5	0
Credit	2	17	9	67	2	0	1	1
Inflation	7	22	18	2	18	1	5	8
FFR	14	28	33	3	2	1	4	2
Hours	10	17	20	2	26	2	2	4
Wages	1	2	2	0	11	0	0	4
	Model with long-term debt and $\psi = 0$							
GDP	9	55	5	4	16	2	6	3
Investment	9	54	9	4	17	0	6	1
Consumption	4	15	24	1	22	1	1	7
Spread	9	21	24	2	38	0	1	3
Equity	12	4	9	51	17	0	4	2
Credit	12	11	45	13	16	1	1	1
Inflation	6	29	25	1	28	0	3	4
FFR	6	34	23	2	29	0	2	2
Hours	2	17	14	1	58	1	2	5
Wages	2	13	7	1	19	0	1	3

Note: This table reports the variance decomposition of the observables used in the estimation. The decomposition is computed at the posterior mode for the estimated parameters. The column labeled "Risk" combines the contribution of the unanticipated and anticipated risk shocks. "M.E.I." stands for marginal efficiency of investment shock, "NW" stands for net worth shock, and "M.P." stands for monetary policy shock.

Supplemental Appendix

Corporate Debt Maturity and Business Cycle Fluctuations

Francesco Ferrante, Andrea Prestipino, and Immo Schott

A Equilibrium equations

This section reports the equilibrium equations of our baseline model.

- Workers

$$1 = \mathbb{E}_t \Lambda_{t+1} \frac{R_t^d}{\pi_{t+1}} \tag{1}$$

$$1 = \mathbb{E}_t \Lambda_{t+1} \frac{R_t}{\pi_{t+1}} \tag{2}$$

where $\Lambda_{t+1} = \beta \frac{U_c(t+1)}{U_c(t)}$ is the stochastic discount factor with the marginal utility of consumption given by

$$U_c(t) = \frac{\xi_t^c}{C_t - hC_{t-1}} - \beta h \mathbb{E}_t \frac{\xi_{t+1}^c}{C_{t+1} - hC_t}.$$

- Entrepreneurs

$$1 = \Psi'(\omega_t)\, \gamma_t \tag{3}$$

$$\begin{aligned} &\left(Q_t^l + \frac{dQ_t^l}{d\eta_t}\eta_t\right)\gamma_t - \frac{(1-\lambda_l)\,\eta_{t-1}}{\pi_t}\frac{dQ_t^l}{d\eta_t}\frac{Q_t^k - Q_t^l\eta_t}{\Psi(\omega_t)\,\mu_t}\varphi_t \\ &\quad = \mathbb{E}_t \Lambda_{t,t+1} \frac{c_l + (1-\lambda_l)\,Q_{t+1}^l}{\pi_{t+1}} \varphi_{t+1} \left(1 - F_t\left(\bar{\xi}_{t+1}, \sigma_t\right)\right) \end{aligned} \tag{4}$$

$$X_t = \Psi(\omega_t) \left(\tilde{\xi}_t - \bar{\xi}_t\right) R_t^k Q_{t-1}^k K_{t-1} \left(1 - F\left(\bar{\xi}_t^l\right)\right) + T_t^e + \xi_t^n, \tag{5}$$

$$K_t = \frac{X_t}{Q_t^k - \eta_t Q_t^l}. \tag{6}$$

where

$$R_t^k = \frac{r_t^k + (1-\delta)Q_t^k}{Q_{t-1}^k}, \tag{7}$$

$$R_t^l = \frac{1}{\pi_t} \cdot \frac{c_l + (1-\lambda_l)\, Q_t^l}{Q_{t-1}^l}. \tag{8}$$

$$\Psi(\omega_t) = (1-\overline{\psi})\omega_t - \frac{\psi}{2}(\omega_t - \overline{\omega})^2. \tag{9}$$

$$\gamma_t = \frac{1}{Q_t^k - Q_t^l \eta_t} \mathbb{E}_t \Lambda_{t,t+1} \varphi_{t+1} \int_{\bar{\xi}_{t+1}} \left(\xi_{t+1} - \bar{\xi}_{t+1}\right) R_{t+1}^k Q_t^k dF_t\left(\xi_{t+1}, \sigma_t\right) \tag{10}$$

$$\varphi_t = \left\{(1-\omega_t) + \Psi\left(\omega_t\right)\gamma_t\right\} \tag{11}$$

$$\bar{\xi}_{t+1} = \frac{R_{t+1}^l Q_t^l}{R_{t+1}^k Q_t^k} \eta_t \tag{12}$$

$$\mu_t = \tilde{\xi}_t R_t^k Q_{t-1}^k - \frac{c_l + (1-\lambda_l)\, Q_t^l}{\pi_t} \eta_{t-1}, \tag{13}$$

$$\tilde{\xi}_t = \frac{1}{1 - F_{t-1}\left(\bar{\xi}_t\right)} \int_{\bar{\xi}_t} \xi_t dF_{t-1}\left(\xi_t\right) \tag{14}$$

$$F_{t-1}(\bar{\xi}_t) = \Phi\left(\frac{log(\bar{\xi}_t) + .5(\sigma_{t-1})^2}{\sigma_{t-1}^e}\right) \tag{15}$$

$$\int_{\bar{\xi}_t} \xi_t dF_{t-1}\left(\xi_t\right) = 1 - \Phi\left(\frac{log(\bar{\xi}_t) - .5(\sigma_{t-1})^2}{\sigma_{t-1}}\right) \tag{16}$$

and $\Phi()$ is the cdf of a standard Normal distribution.

- Financial intermediary

$$Q_t^l = \mathbb{E}_t \Lambda_{t+1} \kappa_t \left\{ \left[1 - F_t\left(\bar{\xi}_{t+1}, \sigma_t\right)\right] \frac{c_l + (1-\lambda_l) Q_{t+1}^l}{\pi_{t+1}} + \gamma_l \frac{Q_t^k R_{t+1}^k}{\eta_t} \int_0^{\bar{\xi}_{t+1}} \xi_{t+1} dF_t\left(\xi_{t+1}, \sigma_t\right) \right\} \tag{17}$$

$$\frac{\partial Q_t^l}{\partial \eta_t} = -\mathbb{E}_t \Lambda_{t+1} \kappa_t \left\{ \frac{\partial \bar{\xi}_{t+1}}{\partial \eta_t} f_t\left(\bar{\xi}_{t+1}\right) \left[Q_t^l R_{t+1}^l - \gamma_l \frac{Q_t^k R_{t+1}^k}{\eta_t} \bar{\xi}_{t+1} \right] + \gamma_l \frac{Q_t^k R_{t+1}^k}{(\eta_t)^2} \int_0^{\bar{\xi}_{t+1}} \xi_{t+1} dF_t\left(\xi_{t+1}\right) \right.$$
$$\left. + \mathbb{E}_t \Lambda_{t+1} \kappa_t [1 - F_t\left(\bar{\xi}_{t+1}\right)] (1-\lambda_l) \frac{\partial Q_{t+1}^l}{\partial \eta_{t+1}} \eta_{\eta,t+1}\left(\eta_t\right) \right. \tag{18}$$

- Intermediate goods producers

$$Y_t^m = A_t (u_t K_{t-1})^{\chi} (H_t)^{1-\chi}, \tag{19}$$

$$w_t = p_t^m (1-\chi) \frac{Y_t}{H_t} \tag{20}$$

$$r_t^k = p_t^m \chi \frac{Y_t}{K_{t-1}} \tag{21}$$

$$\frac{r_t^k}{u_t} = a'(u_t). \tag{22}$$

$$a(u_t) = \bar{\gamma}_u (e^{\gamma_u (u_t - 1)} - 1) \tag{23}$$

- Retailers

$$\left(\frac{\pi_t}{\tilde{\pi}_t} - 1\right) \frac{\pi_t}{\tilde{\pi}_t} = \left[P_t^m \frac{\varepsilon_t}{(\varepsilon_t - 1)} - 1 \right] \gamma_p + \beta \mathbb{E}_t \Lambda_{t,t+1} \left(\frac{\pi_{t+1}}{\tilde{\pi}_{t+1}} - 1\right) \frac{\pi_{t+1}}{\tilde{\pi}_{t+1}} \frac{Y_{t+1}}{Y_t} \tag{24}$$

- Labor unions

$$\frac{\pi_t^w}{\tilde{\pi}_t^w} \left(\frac{\pi_t^w}{\tilde{\pi}_t^w} - 1\right) = \left(\frac{\bar{\varphi} H_t^{\varphi}}{U_{ct}} \frac{\varepsilon_t^w}{\varepsilon_t^w - 1} - W_t \right) \gamma_w + \beta \mathbb{E}_t \Lambda_{t,t+1} \frac{\pi_{t+1}^w}{\tilde{\pi}_{t+1}^w} \left(\frac{\pi_{t+1}^w}{\tilde{\pi}_{t+1}^w} - 1\right) \frac{H_{t+1}}{H_t}, \tag{25}$$

- Capital goods producers

$$Q_t^k = \mu_t^k \left(1 + \frac{\gamma_k}{2}\left(\frac{I_t}{I_{t-1}} - 1\right)^2 + \gamma_k \frac{I_t}{I_{t-1}}\left(\frac{I_t}{I_{t-1}} - 1\right)\right) - \beta \mathbb{E}_t \Lambda_{t+1} \mu_{t+1}^k \gamma_k \left(\frac{I_{t+1}}{I_t}\right)^2 \left(\frac{I_{t+1}}{I_t} - 1\right) \tag{26}$$

$$K_t = (1-\delta)K_{t-1} + I_t. \tag{27}$$

- Monetary policy

$$\log(R_t) = (1-\rho_r)\, log(R_{SS}) + \rho_r \log(R_{i,t-1}) + (1-\rho_r)\left(\kappa_\pi \log\left(\frac{\pi_t}{\bar{\pi}_t}\right) + \kappa_y \log\left(\frac{Y_t}{Y_{t-1}}\right)\right) + \xi_t^m. \tag{28}$$

- Goods market clearing

$$Y_t - \nu\left[K_{t-1}(1-\gamma_l)\int_0^{\bar{\xi}_t} \xi_t dF_t(\xi_t)\right] = C_t + I_t + G_t + a(u_t)K_{t-1}, \tag{29}$$

B Model without equity adjustment costs

In this section, we show that in a model with defaultable long-term debt and costless equity issuance the equilibrium is not well-defined. By setting $\psi = 0$, we can rewrite the value function of the entrepreneur as

$$\begin{aligned} v_t(\eta_{t-1}, \xi_t) &= \max_{\eta_t, \omega_t} \left\{ \mu_t(\eta_t, \eta_{t-1}, \xi_t)(1-\omega_t) + \mu_t(\eta_t, \eta_{t-1}, \xi_t) \frac{\omega_t\left(1-\bar{\psi}\right)}{Q_t^k - Q_t^l \eta_t} \mathbb{E}_t \Lambda_{t,t+1} \max\{0, v_t(\eta_t, \xi_{t+1})\} \right. \\ &= \mu_t(\eta_t, \eta_{t-1}, \xi_t) + \omega_t g_t(\eta_t) \end{aligned}$$

where

$$\mu_t(\eta_t, \eta_{t-1}, \xi_t) = \frac{n_t}{k_{t-1}} = \left[\xi_t R_t^k q_{t-1}^k - \frac{\left(c_l + (1-\lambda_l) Q_t^l\right)}{\pi_t} \eta_{t-1}\right] \tag{30}$$

$$g_t(\eta_t) = \mu_t(\eta_t, \eta_{t-1}, \xi_t)\left(-1 + \frac{\left(1-\bar{\psi}\right)\mathbb{E}_t \Lambda_{t,t+1} \max\{0, v_t(\eta_t, \xi_{t+1})\}}{Q_t^k - Q_t^l \eta_t}\right) \tag{31}$$

where $g_t(\eta_t)$ represents the net return on an extra unity of equity inside the firm.

The first order conditions for equity and leverage, ω_t^* and η_t^*, are

$$g_t(\eta_t^*) = 0 \tag{32}$$

$$\frac{d\mu_t}{d\eta_t^*} + \omega_t^* \frac{dg_t(\eta_t^*)}{d\eta_t^*} = 0 \tag{33}$$

where

$$\frac{d\mu_t}{d\eta_t^*} = -(1-\lambda_l)\frac{dQ_t^l}{d\eta_t}\frac{\eta_{t-1}}{\pi_t} > 0 \tag{34}$$

because $\frac{dQ_t^l}{d\eta_t} < 0$. However, these first order conditions necessarily select a saddle point. In fact, equations (33) and (34) imply that $\frac{dg_t(\eta_t^*)}{d\eta_t^*} < 0$. If this is the case, then we can find another ε such that $\hat{\eta}_t = \eta_t^* - \varepsilon$ and $g_t(\hat{\eta}_t) > 0$, and set ω_t to infinity to yield an infinite value for the entrepreneur. The entrepreneur could reduce leverage to increase the excess return on equity and then freely adjust equity to infinity to obtain an infinite return. Hence, without some increasing marginal costs of adjusting equity, the entrepreneur's problem does not have an interior optimum and consequently the macroeconomic model does not have an equilibrium. Quadratic costs of equity adjustment alter the first order conditions by making the return on equity depend on ω_t, that is $g_t(\eta_t, \omega_t)$, preventing the entrepreneur from pursuing the strategy described above.

On the other hand, if debt is short term, that is if $\lambda = 1$, an interior optimum would be possible even with $\psi = 0$. In fact, using $\frac{d\mu_t}{d\eta_t^*} = 0$, the first order conditions would become

$$g_t\left(\eta_t^{*,ST}\right) = 0 \tag{35}$$

$$\frac{dg_t\left(\eta_t^{*,ST}\right)}{d\eta_t^{*,ST}} = 0. \tag{36}$$

In this case, equation (36) implies that the optimal leverage is chosen in order to maximize the expected return on equity, preventing any profitable deviation from the interior optimum.

This analysis suggests that in a model with defaultable long-term debt some form of equity adjustment cost is a necessary condition for the existence of an equilibrium. We can think of the model with $\psi = 0$ from Section 3 as a limit case with arbitrarily small equity adjustment costs.

An additional result for the model with $\psi = 0$ is that the derivative of the entrepreneurial leverage policy function is zero, that is $\eta_\eta = 0$. This can be seen by rewriting equation (32) as

$$\frac{1}{\left(1-\bar{\psi}\right)} = \frac{\mathbb{E}_t \Lambda_{t,t+1} \max\left\{0, v_t\left(\eta_t, \xi_{t+1}\right)\right\}}{Q_t^k - Q_t^l \eta_t} \tag{37}$$

which is equivalent to equation (36) in the paper. In this case, equation (37) pins down η_t as a function of only the aggregate price of capital and the household stochastic discount factor. Leverage η_t will not depend on past leverage η_{t-1} or ω_t and hence $\eta_\eta = 0$. Equation (33) will instead determine ω_t.

C Algorithm to compute derivative of leverage

In our baseline model, the entrepreneur's individual choice of equity (ω_t^i) and leverage (η_t^i) is characterized by the following first order conditions

$$1 = \Psi'\left(\omega_t^i\right)\gamma_t^i \tag{38}$$

$$\left(Q_t^{l,i} + \frac{dQ_t^{l,i}}{d\eta_t^i}\eta_t^i\right)\gamma_t^i - \frac{(1-\lambda_l)\eta_{t-1}^i}{\pi_t}\frac{dQ_t^{l,i}}{d\eta_t^i}\frac{Q_t^k - Q_t^{l,i}\eta_t^i}{\Psi\left(\omega_t^i\right)\mu_t^i}\varphi_t^i$$
$$= \tag{39}$$
$$\mathbb{E}_t\Lambda_{t,t+1}\frac{c_l+(1-\lambda_l)Q_{t+1}^{l,i}}{\pi_{t+1}}\varphi_{t+1}^i\left(1-F_t\left(\bar{\xi}_{t+1}^i,\sigma_t\right)\right)$$

where

$$Q_t^{l,i} = \mathbb{E}_t\Lambda_{t+1}\kappa_t\left\{\left[1-F_t\left(\bar{\xi}_{t+1}^i,\sigma_t\right)\right]\frac{c_l+(1-\lambda_l)Q_{t+1}^{l,i}}{\pi_{t+1}} + \gamma_l\frac{Q_t^k R_{t+1}^k}{\eta_t^i}\int_0^{\bar{\xi}_{t+1}^i}\xi_{t+1}dF_t\left(\xi_{t+1},\sigma_t\right)\right\} \tag{40}$$

$$\frac{\partial Q_t^{l,i}}{\partial\eta_t^i} = -\mathbb{E}_t\Lambda_{t+1}\kappa_t\left\{\frac{\partial\bar{\xi}_{t+1}^i}{\partial\eta_t^i}f_t\left(\bar{\xi}_{t+1}^i\right)\left[Q_t^{l,i}R_{t+1}^{l,i} - \gamma_l\frac{Q_t^k R_{t+1}^k}{\eta_t^i}\bar{\xi}_{t+1}^i\right] + \gamma_l\frac{Q_t^k R_{t+1}^k}{\left(\eta_t^i\right)^2}\int_0^{\bar{\xi}_{t+1}^i}\xi_{t+1}dF_t\left(\xi_{t+1}\right)\right\}$$
$$+\mathbb{E}_t\Lambda_{t+1}\kappa_t[1-F_t\left(\bar{\xi}_{t+1}^i\right)](1-\lambda_l)\frac{\partial Q_{t+1}^{l,i}}{\partial\eta_{t+1}^i}\eta_{\eta,t+1}\left(\eta_t^i\right) \tag{41}$$

$$\gamma_t^i = \frac{1}{Q_t^k - Q_t^{l,i}\eta_t^i}\mathbb{E}_t\Lambda_{t,t+1}\varphi_{t+1}^i\int_{\bar{\xi}_{t+1}^i}\left(\xi_{t+1}-\bar{\xi}_{t+1}^i\right)R_{t+1}^k Q_t^k dF_t\left(\xi_{t+1},\sigma_t\right) \tag{42}$$

$$\varphi_t^i = \left\{\left(1-\omega_t^i\right) + \Psi\left(\omega_t^i\right)\gamma_t^i\right\} \tag{43}$$

$$\bar{\xi}_{t+1}^i = \frac{\left(c_l+(1-\lambda_l)Q_{t+1}^{l,i}\right)}{\pi_{t+1}}\frac{\eta_t^i}{R_{t+1}^k Q_t^k} \tag{44}$$

$$\mu_t^i = \tilde{\xi}_t R_t^k Q_{t-1}^k - \frac{c_l+(1-\lambda_l)Q_t^{l,i}}{\pi_t}\eta_{t-1}^i, \tag{45}$$

These equations determine a policy function for *individual* leverage which we denote as $\tilde{\eta}^i(\eta_{t-1}^i, S_t, \sigma^\epsilon)$, where η_{t-1}^i is the *individual* leverage at time $t-1$, S_t is the aggregate state of the economy, which also includes the *aggregate* level of leverage at time $t-1$, and σ^ϵ is a vector collecting the standard deviations of all exogenous shocks. Note that this policy function is different from the policy function for *aggregate* leverage.

Our algorithm to approximate $\frac{\partial\tilde{\eta}^i(\eta_{t-1}^i,S_t,\sigma^\epsilon)}{\partial\eta_{t-1}^i}$ works in two steps. In the

first step, we compute $\tilde{\eta}^i(\eta^i_{t-1}, S^{ss}, 0)$ and its derivative $\frac{\partial \tilde{\eta}^i(\eta^i_{t-1}, S^{ss}, 0)}{\partial \eta^i_{t-1}}$ in the steady state. This is done with the following iterative procedure:

- Given an initial guess for S^{ss} and for associated prices and quantities in steady state, and a guess for $\eta_{\eta,ss}$, we use global methods to solve for $\tilde{\eta}^i(\eta^i_{t-1}, S^{ss}, 0)$ over a grid on entrepreneurial leverage η^i_{t-1}.
- Given the policy function $\tilde{\eta}^i(\eta^i_{t-1}, S_{ss}, 0)$, we can compute $\eta_{\eta,SS} = \frac{\partial \tilde{\eta}^i(\eta^i_{t-1}, S_{ss}, 0)}{\partial \eta^i}$ and update $\eta_{\eta,ss}$ and S_{ss}.
- These steps are repeated until convergence.

In the second step of our algorithm we approximate the derivative function denoted by

$$\tilde{g}(\eta^i_{t-1}, S_t, \sigma^\epsilon) = \frac{\partial \tilde{\eta}^i(\eta^i_{t-1}, S_t, \sigma^\epsilon)}{\partial \eta^i_{t-1}}$$

linearly around the steady state. To do so we write the policy function for leverage in an economy without aggregate risk, as an explicit function of previous leverage, aggregate prices at time t, expected prices at $t+1$, and all other aggregate quantities that the entrepreneur takes as given in the system (38)–(45). That is, letting $\Gamma_t = \{\eta^i_{t-1}, \tilde{\xi}_t, \sigma_t, \kappa_t, \mathbb{E}_t R^k_{t+1}, R^k_t, Q^k_t, Q^k_{t-1}, \mathbb{E}_t \Lambda_{t+1}, \mathbb{E}_t \pi_{t+1}, \pi_t\}$ we denote by $\eta(\Gamma_t)$ the policy function that solves the optimality conditions above without aggregate uncertainty and for a level of previous leverage, prices at time t, expected prices at $t+1$, and all other aggregate quantities given by Γ_t. We let $g(\Gamma_t) = \frac{\partial \eta}{\partial \eta_{t-1}}$ and our approximation be given by

$$\hat{g}(\Gamma_t) = g(\Gamma^{ss}) + \nabla g d\Gamma_t$$

where the gradient of g is computed numerically by solving the policy functions for η globally for small deviations of each argument in Γ_t from steady state.

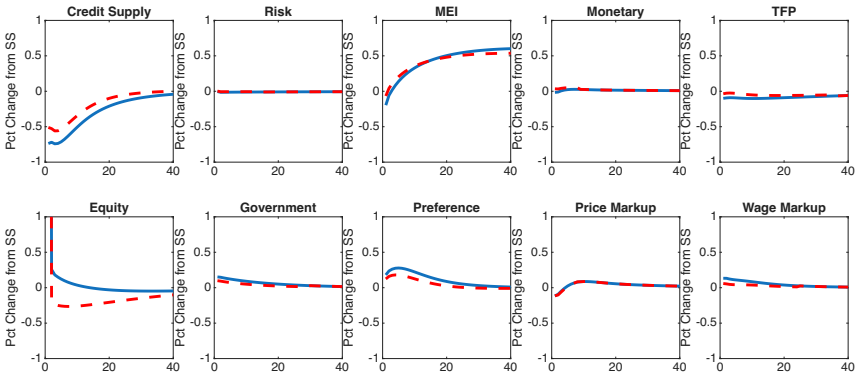


Figure C.1: Accuracy of derivative algorithm

Note: Figure reports the impulse responses of the derivative of the entrepreneurial leverage policy function computed either with our linear approximation method , that is $\hat{g}(\Gamma_t)$ (blue solid line), or computed non-linearly, that is $g(\Gamma_t)$ (red dashed line). In each panel we consider a one standard deviation of the shock, as obtained in our estimated baseline model.

C.1 Accuracy of the algorithm

For sufficiently small shocks, the linear approximation provides an accurate characterization of the behavior of the function $g(\cdot)$. As an additional validation of our computational approach, Figure (C.1) compares the impulse response implied by the linearized policy function, $\hat{g}(\Gamma_t)$ (solid blue line), with the response obtained from the derivative of the policy function computed using global solution methods, $g(\Gamma_t)$ (red dashed line). In both cases, the responses are evaluated along the same linear path for Γ_t following an unanticipated shock. The path of $g(\Gamma_t)$ captures the nonlinear relationship between entrepreneurial leverage and aggregate variables. As shown in Figure (C.1), the two solution methods deliver very similar paths for $g()$, providing support for our linear algorithm.

D Estimation of equity adjustment costs

This appendix provides more detail on the data treatment, estimation, and robustness of the results presented in Section IV.B.

We perform the following cleaning steps on our quarterly Compustat sample (S&P Global Market Intelligence, 2025). We remove firms in the finance, government, and utility sectors. We remove non-U.S. firms. Observations with negative total assets are dropped. We transform cumulative variables `sstky|`, `dvy`, and `prstkcy` into quarterly variables.

Our dependent variable is $R^x_{i,t+1} = \frac{V_{i,t+1}}{X_{i,t}}$, where $V_{i,t+1}$ is the $t+1$ cum-dividend market value of equity. It is constructed using Compustat variables `cshoq`×`prccq` + `netpayouts`, where `netpayouts` = `dvyq` - (`sstkyq` - `prstkcyq`). We remove negative observations of $V_{i,t+1}$. The denominator $X_{i,t}$ is the cum-dividend book value of equity, defined as `ceqq` + `netpayouts`. Again, we remove negative observations.

Our key independent variable is $\omega_{it} = 1 - \frac{\texttt{netpayouts}}{X_{i,t}}$, i.e., one minus the fraction of net payouts to shareholders over the cum-dividend book value of equity. We remove observations of ω_{it} that include negative values of $X_{i,t}$.

Finally, we trim the independent variable at the 5th and 95th percentile. The dependent variable is truncated at zero from below, we therefore only trim it at the 95th percentile.[1]

The regression we run is

$$\frac{R^x_{i,t+1}}{1+r_{t+1}} = \psi\omega_{i,t} + \delta_i + \delta_t + \nu_{it+1}, \tag{46}$$

where r_{t+1} is the quarterly realized real rate and δ_i and δ_t are firm- and time fixed effects.

Our results are shown in Table D.1. The first column shows the regression coefficient of the model estimated in (46). Columns (2) and (3) show the effect of removing the firm or time fixed effects.

[1] Winsorizing instead of trimming the data leads to similar results.

Table D.1: Estimating the equity adjustment cost parameter

	(1)	(2)	(3)	(4)	(5)	(6)
ω	10.85	15.43	11.77	10.74	15.26	11.65
	(0.13)	(0.15)	(0.13)	(0.13)	(0.15)	(0.13)
Firm fixed effects	yes	no	yes	yes	no	yes
Time fixed effects	yes	yes	no	yes	yes	no
Additional cleaning	no	no	no	yes	yes	yes
N	513,257	513,785	513,257	495,070	495,518	495,070

Robustness As a robustness exercise, we perform additional cleaning steps. We only keep firms with at least five quarters of information in the sample. Further, we remove firm-quarters that violate the accounting identity by more than 10 per cent of the book value of assets (as in Covas and Den Haan (2012)). Lastly, we drop firms involved in major mergers (Compustat footnote `saleq_fn1` code "AB"). As indicated in columns (4)–(6) of Table D.1, these additional cleaning steps hardly affect our results.

E Additional model experiments

This appendix provides additional results for our model. Figures E.1 and E.2 show the impulse responses to a debt shock in the baseline model without equity adjustment costs and with infinite equity adjustment costs, respectively, and with no real default costs, i.e. $\nu = 0$. Figure E.3 presents the impulse response to a credit supply shock in a model with a different calibration of the Taylor rule and of the slope of the Phillips curve. Figure E.4 shows the impulse responses to a net worth shock in the baseline model. Figure E.5 compares the impulse responses in the baseline model with those in a model with $\psi = \infty$. Table E.1 shows the variance decomposition for that case. Finally, Figure E.6 compares the impulse responses when debt is short- or long-term in the case of $\psi = \infty$.

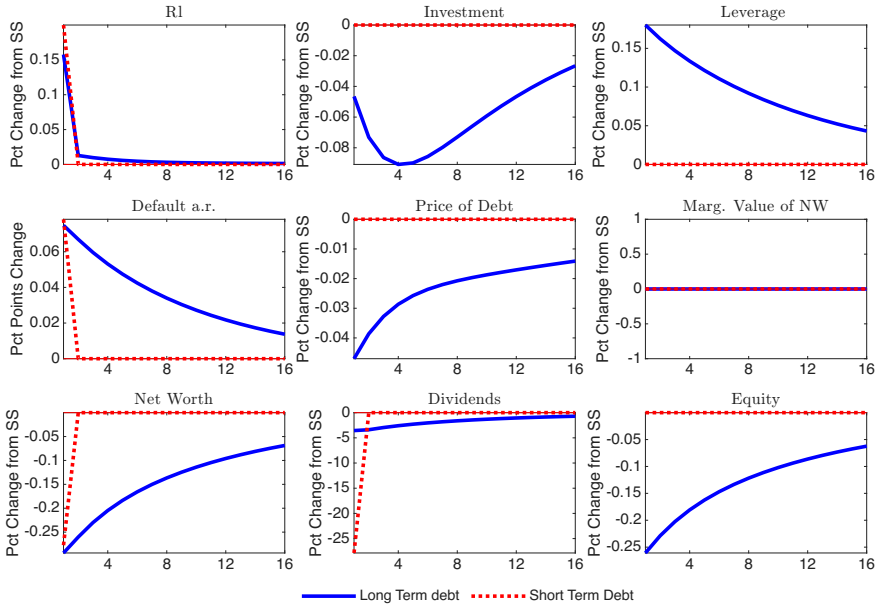


Figure E.1: Debt shock in model with sticky prices: no equity adjustment costs

Note: Figure shows the impulse response to a one-time shock to the value of outstanding debt in a version of the baseline model with no real default costs, i.e. $\nu = 0$.

Table E.1: Variance Decomposition: Model with $\psi = \infty$

Model with long-term debt and $\psi = \infty$								
	Credit Supply	Risk	M.E.I.	NW	Markup	Gov	M.P.	TFP
GDP	47	8	5	6	16	7	3	3
Investment	54	9	13	7	11	0	2	1
Consumption	8	7	24	2	2	1	0	4
Spread	9	68	13	8	0	0	0	0
Equity	18	2	12	64	1	0	1	0
Credit	17	10	37	31	1	1	1	0
Inflation	37	13	8	2	18	1	4	5
FFR	45	19	21	4	1	1	1	1
Hours	28	11	33	2	14	1	1	1
Wages	2	2	4	1	10	0	0	3

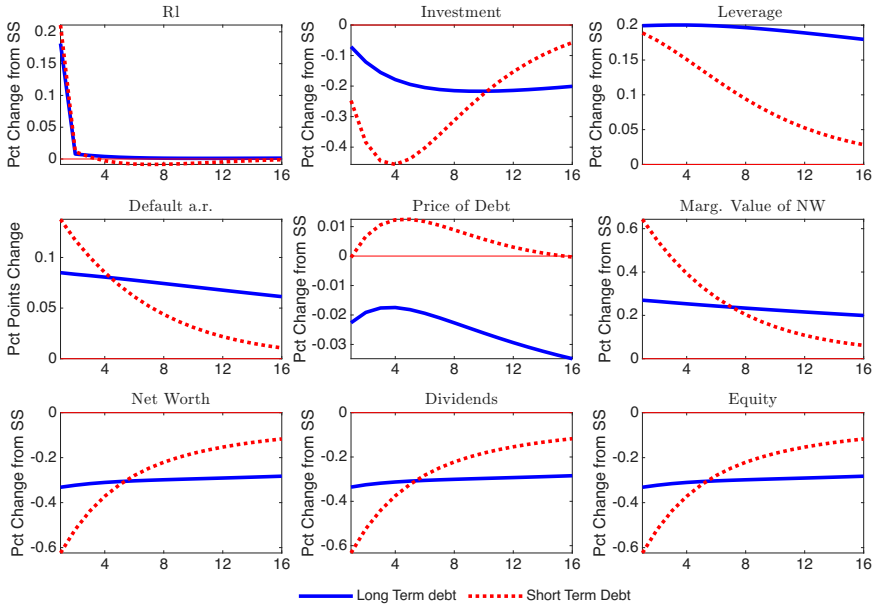


Figure E.2: Debt shock in model with sticky prices: infinite equity adjustment costs

Note: Figure shows the impulse response to a one-time shock to the value of outstanding debt in a version of the baseline model with infinite equity adjustment costs no real default costs, i.e. $\nu = 0$.

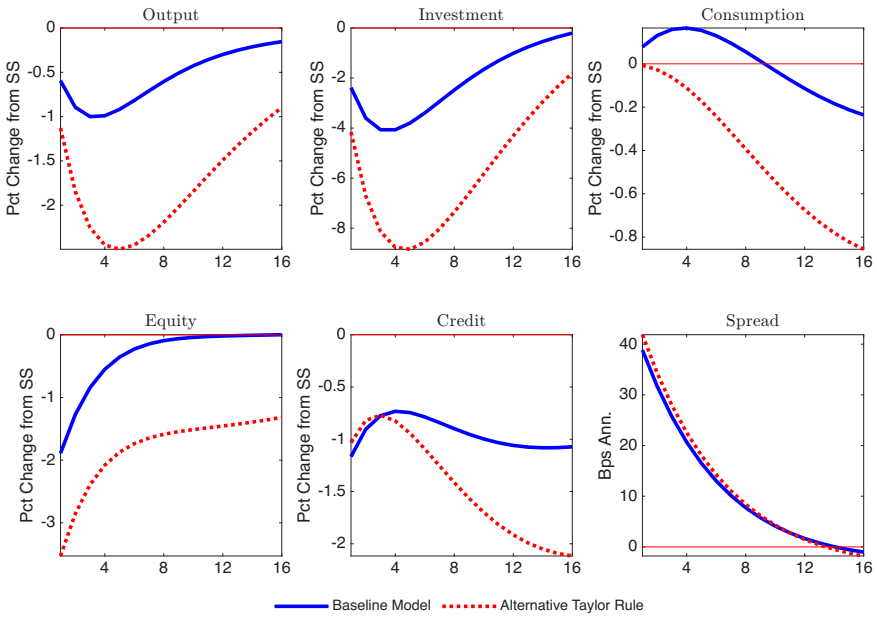


Figure E.3: Credit supply shock in model with alternative calibration

Note: Figure shows the impulse response to a credit supply shock in the baseline model (blue line) and in a version of the model with a lower Phillips curve slope of 2.5 percent and a less aggressive Taylor rule, with a coefficient of 1.5 on inflation and zero on output (red dotted line). The shock is scaled to deliver a 1 percent decline in output in the baseline model.

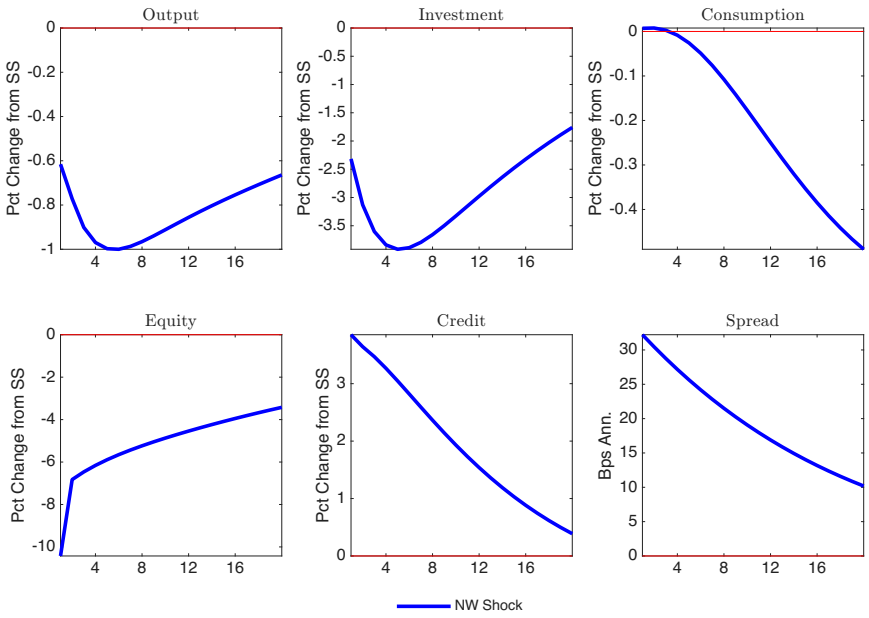


Figure E.4: Net worth shock in baseline model

Note: Figure shows the impulse response to shock to the entrepreneurial net worth in our baseline model

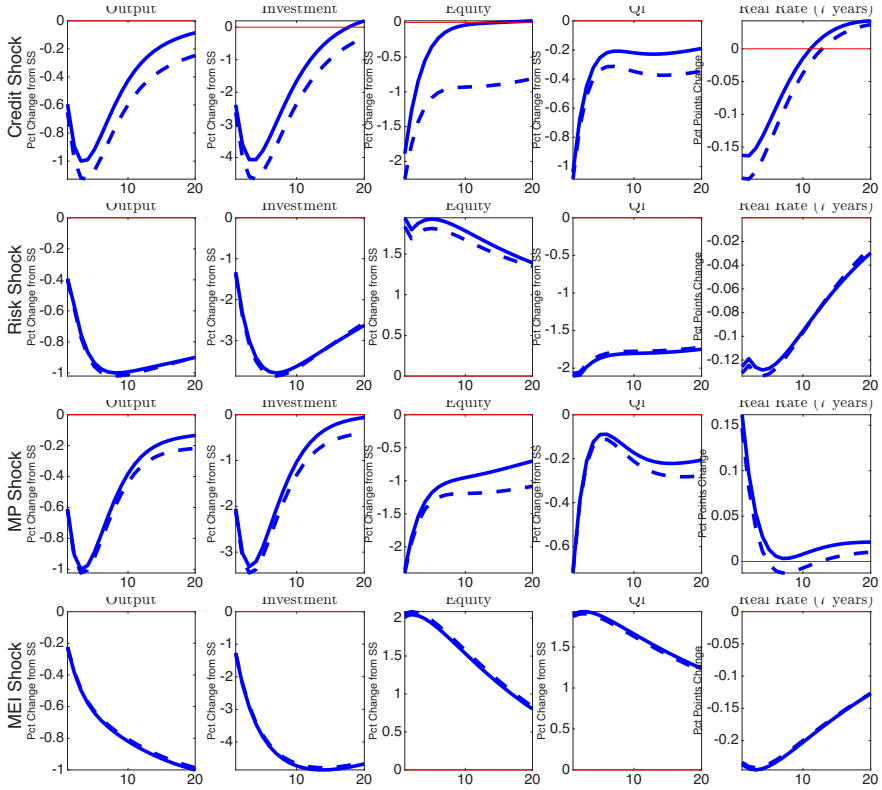


Figure E.5: Impulse responses in baseline model vs model with $\psi = \infty$

Note: Figure compares the impulse responses in our baseline model (blue solid line) to a model with the same parameters but with $\psi = \infty$ (dark blue dashed line).

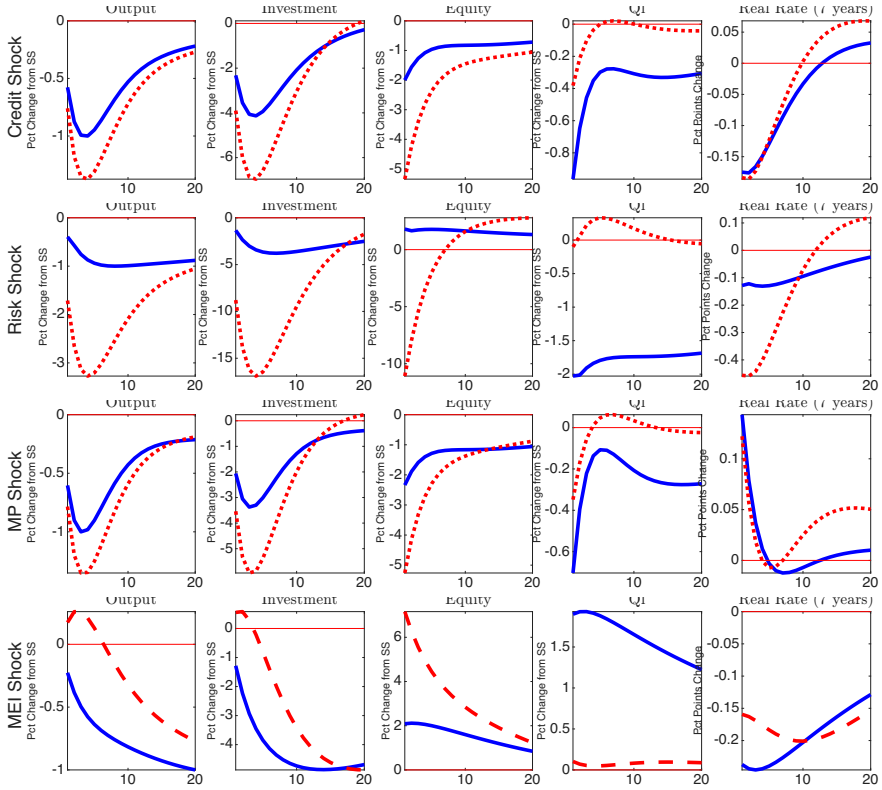


Figure E.6: Long-term debt vs short-term debt in baseline model with $\psi = \infty$

Note: Figure compares the impulse responses in our baseline model (blue solid line) and in a model with short-term debt (red dashed line), in both cases assuming that $\psi = \infty$.

F Financially constrained intermediaries

This section presents additional details on our extension of the baseline model discussed in Section V. We follow Gertler and Karadi (2011) and assume that, at each point in time, a fraction b of household members are bankers managing a financial intermediary. At the beginning of each period, a fraction $(1-\sigma_b)$ of bankers returns to the family and is replaced by an equal number of new bankers, so that the relative proportion of bankers and non-bankers within the family remains constant. New bankers enter with some startup funds from the family, as we discuss below.

A financial intermediary uses its own net worth, N_t^b, and deposits to invest in loans:

$$Q_t^l L_t = N_t^b + D_t \tag{47}$$

A banker's net worth is equal to the difference between the return on assets and the return on liabilities:

$$N_t^b = R_{t+1}^b Q_t^l L_t - \frac{R_t^d}{\pi_{t+1}} D_t \tag{48}$$

The banker's objective is to maximize the payouts to the representative family upon exit, which can be written recursively as

$$V_t^b = E_t \Lambda_{t+1} \left[(1-\sigma_b) N_{t+1}^b + \sigma_b V_{t+1}^b \right] \tag{49}$$

where Λ_{t+1} is the household stochastic discount factor between time t and t+1.

We assume that bankers can divert a portion θ^l of their loans and stop operating their financial subsidiary. As a result, depositors will limit the amount they lend to bankers to ensure that bankers do not have an incentive to divert funds:

$$V_t^b \geq \theta^l Q_t^l L_t. \tag{50}$$

The problem of the banker will be to choose L_t, D_t and N_t^b in order to optimize (49) subject to (47), (48) and (50). Given the linearity of the problem it can be shown that the value function satisfies $V_t^b = \psi_t N_t^b$ where ψ_t represents the bank's franchise value, that is the ratio between the marginal value of wealth inside the bank and the marginal value of wealth to the household. We will show that this variable does not depend on bank specific characteristic.

If we define bank leverage as $\phi_t^b = Q_t^l L_t / N_t^b$ we can then rewrite the bank problem as

$$\max_{\phi_t^b} E_t \Lambda_{t+1} \left[(1-\sigma_b) + \sigma_b \psi_{t+1}\right] N_t^b \left[\phi_t^b \left(R_{t+1}^b - \frac{R_t^d}{\pi_{t+1}}\right) + \frac{R_t^d}{\pi_{t+1}}\right] \tag{51}$$

subject to

$$E_t \Lambda_{t+1} \left[(1-\sigma_b) + \sigma_b \psi_{t+1}\right] \left[\phi_t^b \left(R_{t+1}^b - \frac{R_t^d}{\pi_{t+1}}\right) + \frac{R_t^d}{\pi_{t+1}}\right] \geq \theta^l \phi_t^b. \tag{52}$$

In our calibration the incentive constraint is binding and the optimality condition for leverage is given by the incentive constraint at equality:

$$E_t \Omega_{t+1} \left[\phi_t^b \left(R_{t+1}^b - \frac{R_t^d}{\pi_{t+1}}\right) + \frac{R_t^d}{\pi_{t+1}}\right] = \theta^l \phi_t^b, \tag{53}$$

where $\Omega_t = \Lambda_t \left[(1-\sigma_b) + \sigma_b \psi_t\right]$. We can rewrite the optimality condition of leverage in (53) as:

$$E_t \Lambda_{t+1}^b R_{t+1}^b = 1, \tag{54}$$

where the banks stochastic discount factor, Λ_{t+1}^b, is:

$$\Lambda_{t+1}^b = \frac{\Omega_{t+1}}{\theta^l + E_t \Omega_{t+1} \frac{R_t^d}{\pi_{t+1}} \left(\frac{\phi_t^b - 1}{\phi_t^b}\right)} \tag{55}$$

Hence, we can replace the loan pricing equations in the model with

$$Q_t^l = \mathbb{E}_t \Lambda_{t+1}^b \left\{ \left[1 - F_t\left(\bar{\xi}_{t+1}, \sigma_t\right)\right] \frac{c_l + (1-\lambda_l) Q_{t+1}^l}{\pi_{t+1}} + \gamma_l \frac{Q_t^k R_{t+1}^k}{\eta_t} \int_0^{\bar{\xi}_{t+1}} \xi_{t+1} dF_t\left(\xi_{t+1}, \sigma_t\right) \right\} \tag{56}$$

and

$$\frac{\partial Q_t^l}{\partial \eta_t} = -\mathbb{E}_t \Lambda_{t+1}^b \left\{ \frac{\partial \bar{\xi}_{t+1}}{\partial \eta_t} f_t\left(\bar{\xi}_{t+1}\right) \left[Q_t^l R_{t+1}^l - \gamma_l \frac{Q_t^k R_{t+1}^k}{\eta_t} \bar{\xi}_{t+1} \right] + \gamma_l \frac{Q_t^k R_{t+1}^k}{(\eta_t)^2} \int_0^{\bar{\xi}_{t+1}} \xi_{t+1} dF_t\left(\xi_{t+1}\right) \right\}$$
$$+ \mathbb{E}_t \Lambda_{t+1}^b [1 - F_t\left(\bar{\xi}_{t+1}\right)] (1-\lambda_l) \frac{\partial Q_{t+1}^l}{\partial \eta_{t+1}} \eta_{\eta,t+1}\left(\eta_t\right) \tag{57}$$

Finally, to close the model, we need to specify aggregate credit supply and the evolution of bankers net worth as:

$$\phi_t^b = \frac{Q_t^l L_t}{N_t^b}, \tag{58}$$

$$N_t^b = \sigma_b \left\{ R_t^{l,b} Q_{t-1}^l L_{t-1} - \frac{R_{t-1}^d}{\pi_t} D_t \right\} + T^b N_{t-1}^b \tag{59}$$

where T^b controls the transfer to new bankers.

The results in Section V are obtained by calibrating the extended model as follows. We set $\sigma_b = 0.97$ as in Gertler and Karadi (2011), and we calibrate the parameters $T^b = .0004$ and $\theta^l = .52$ to obtain a steady state bank leverage of 10 and a liquidity premium of 1 percent. These are standard values in the literature, see, for example Gertler, Kiyotaki and Prestipino (2020). All other parameters are the same as in the baseline model.

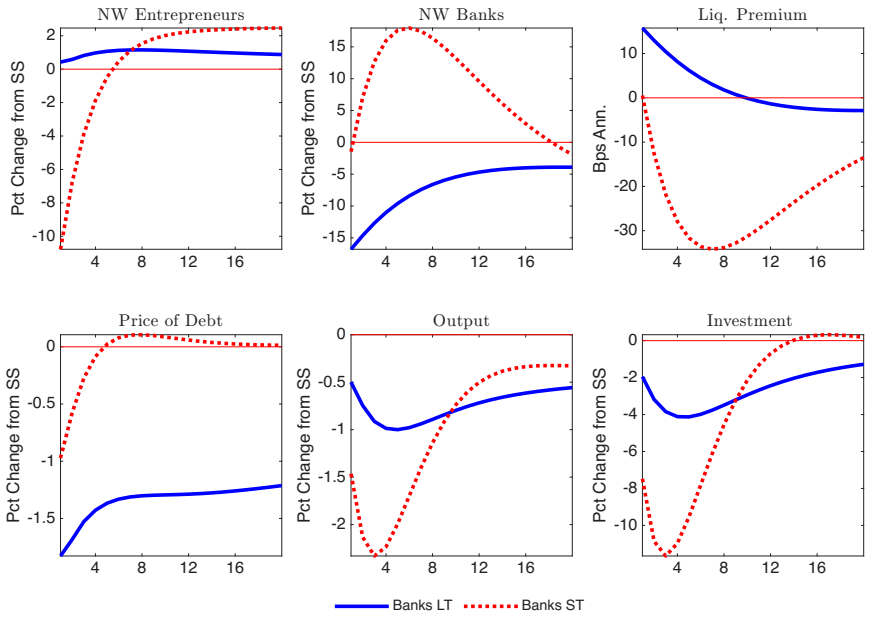


Figure F.7: Risk shock in model with constrained intermediaries

Note: Figure reports the impulse responses to a risk shock in a model extension with constrained financial intermediaries. The blue line represents a model with long term debt and the red dotted line represents a model with short term debt. The shock is calibrated to deliver a 1 percent decline in GDP in the model with long term debt.

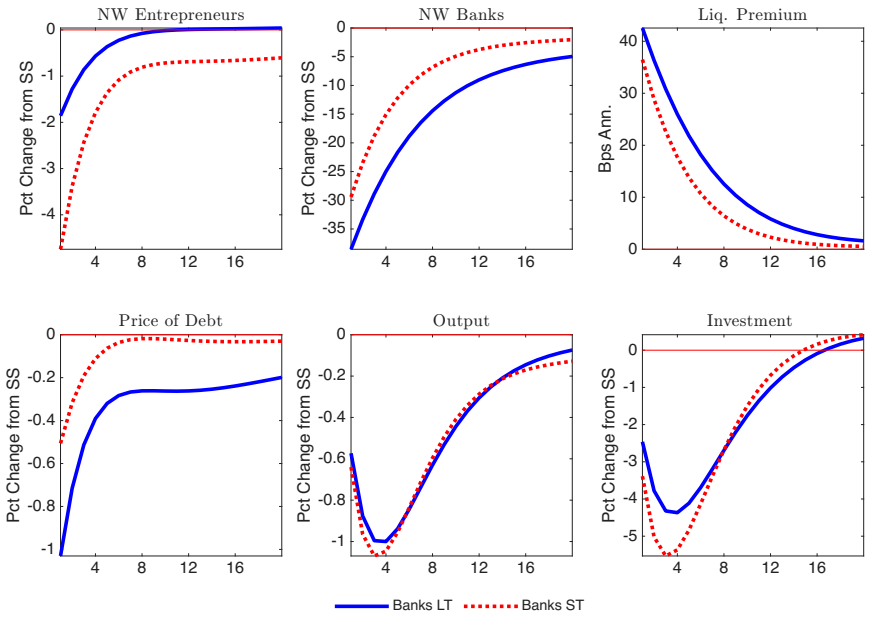


Figure F.8: Bank net worth shock in model with constrained intermediaries

Note: Figure reports the impulse responses to a bank net worth shock in a model extension with constrained financial intermediaries. The blue line represents a model with long term debt and the red dotted line represents a model with short term debt. The shock is calibrated to deliver a 1 percent decline in GDP in the model with long term debt.

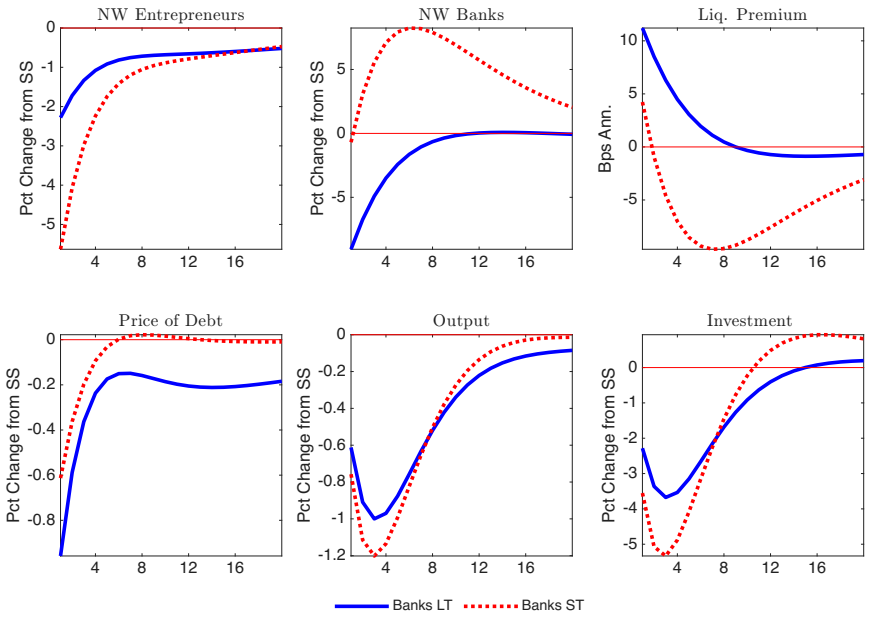


Figure F.9: Monetary shock in model with constrained intermediaries

Note: Figure reports the impulse responses to a monetary policy shock in a model extension with constrained financial intermediaries. The blue line represents a model with long term debt and the red dotted line represents a model with short term debt. The shock is calibrated to deliver a 1 percent decline in GDP in the model with long term debt.

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